Software Engineering (Sessional) Report

Course: CSE-356



CUET Thesis & Project Portal

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Introduction

The **CUET Thesis & Project Portal** is a web-based platform designed to establish a centralized repository for preserving and sharing thesis projects carried out at Chittagong University of Engineering & Technology (CUET). The portal aims to serve as a comprehensive repository for published student theses and projects, and works by alumni. This initiative will assist students in exploring diverse thesis topics, learning from existing works, and discovering new research ideas, thereby fostering academic excellence and collaboration.

1.1 Goals and Objectives of the Project

The primary goals and objectives of the **CUET Thesis & Project Portal** are as follows:

- To create a centralized platform for storing and accessing CUET thesis and projects.
- To give students the opportunity to investigate a variety of thesis subjects and gain knowledge from earlier research.
- To facilitate collaboration between students, faculties, and alumni.
- To encourage academic excellence by showcasing exemplary thesis and projects.

1.2 Scope of the Work

The **CUET Thesis & Project Portal** answers the need for a centralized repository for theses and provides tools for students and faculty to engage in academic research effectively.

1.2.1 Current Situation and Context

Currently, thesis and projects at CUET are stored in fragmented formats, often limited to physical copies or localized digital storage. This lack of centralized access restricts the visibility and usability of these academic contributions, making it difficult for students and faculty to explore existing researchs and projects effectively.

1.2.2 Competing Products (Available in the Market)

There are several online repositories, such as ProQuest and institutional thesis repositories like The Thesis and Project Proposal System of BUET (https://casrproposal.buet.ac.bd/general/proposal), The Thesis of Indian Institute of Technology Delhi (https://web.iitd.ac.in/ sbhalla/thesis.html), and MIT's DSpace (https://dspace.mit.edu/). However, these platforms are either too broad or inaccessible to CUET students due to subscription costs and lack of institutional focus. The **CUET Thesis & Project Portal** will offer a customized solution that is suited to the academic and research requirements of the university.

1.3 System Overview

The **CUET Thesis & Project Portal** will feature:

- A searchable and categorized database of thesis and projects.
- Tools for submitting and updating thesis and project records.
- User-friendly interfaces for students, faculties, and administrators.

1.4 Structure of the Document

The document is structured as follows:

1. **Introduction:** This section provides an overview of the project, including the goals, objectives, scope of work, and the current situation and competing products. It also introduces the system overview and outlines the structure of the document.
2. **Project Management Plan:** This section details the project organization, the life cycle model used, resource requirements, project timeline, and potential social, cultural, and environmental impacts.
3. **Requirement Specifications:** This section identifies system stakeholders, includes graphical and textual descriptions of use cases, and outlines various requirement specifications.

Project Management Plan

2.1 Project Organization

Effective project organization is a crucial first step in software development. It's about ensuring that each team member takes on tasks that match their skills and strengths, allowing everyone to contribute their best. For our project, we have a three-person team, with responsibilities distributed strategically. Each member is assigned tasks that align with their field of expertise and the time required for each project phase, maximizing efficiency and collaboration.

2.1.1 Individual Contribution to the project

Table 2.1: Individual Contribution to the project

Member Name	Requirement Specification	Planning	Designing	Model Building	User Interface	Testing	Deployment
Ratnajit Dhar	✓	✓	✓	✓		✓	✓
Faozia Fariha	✓	✓		✓	✓	✓	
Arpita Mallik	✓	✓	✓		✓	✓	

2.2 Process Model Used

For our project, we have chosen the **Prototyping Model** as the process model. The Prototyping Model emphasizes creating early working versions of the system to gather user feedback. It allows for iterative refinement based on user input, making it suitable for projects where requirements may evolve. This approach enables flexibility, enhances user involvement, and ensures the final product meets user expectations.

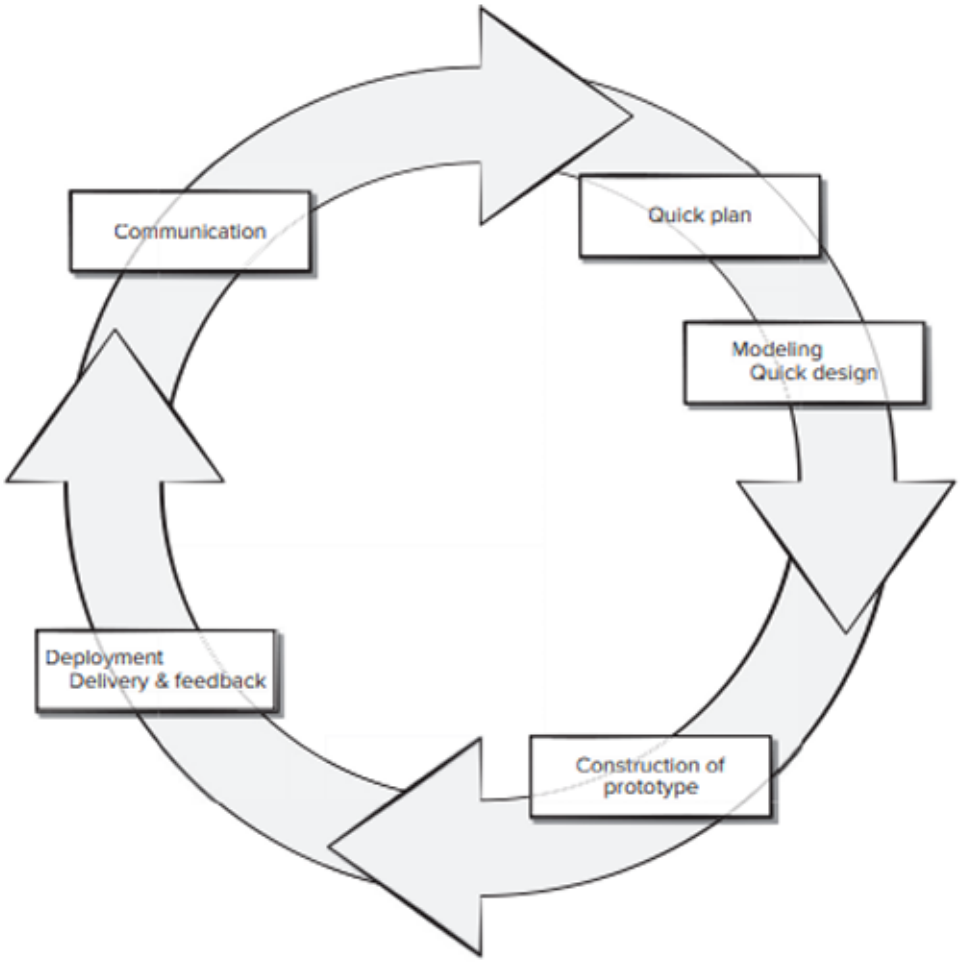


Figure 2.1: Prototyping Process Model

2.2.1 Rationale for Choosing the Prototyping Model

- **Early Problem Detection:** Issues can be identified and resolved at initial stages, minimizing the risk of significant errors later.
- **Continuous Feedback:** Regular input from users at each phase enables improvements based on evolving needs. Continuous involvement of users ensures the system meets expectations and provides a better user experience.
- **Risk Management:** Early identification of risks allows for timely interventions, reducing the likelihood of costly adjustments.
- **Flexibility for Changes:** The iterative nature of prototyping allows changes to be made easily during the development phase, especially when stakeholders request new features or modifications based on the prototype.
- **Improved Communication:** Prototypes enhance understanding between stakeholders and developers, promoting clearer communication and alignment.

2.3 Constraints to Project Implementation

While developing the project, we have identified several constraints that may impact the overall implementation:

- **Schedule Constraints:** Due to academic deadlines, the project must be completed within a limited timeframe. This may affect the depth of research or the time available for testing and refinement.
- **Budget Constraints:** Unforeseen expenses, such as additional hardware requirements or the need for a faster and more responsive server, could strain the project budget, impacting its financial feasibility.
- **Hardware and Software Constraints:**
  - **Software Constraints:** Customizing the existing project extensively may cause delays and raise costs. It's important to balance these needs with the timeline and budget to avoid issues.
  - **Hardware Constraints:** Limitations in network bandwidth or hardware resources may affect the system's performance, scalability, and responsiveness.

2.4 Hardware and Software Resource (Tools/Language) Requirements

The development of the online web app for Research Portal will involve using specific tools and languages.

Languages:

- **Front-End:**
  - The primary framework for front-end development is React.js.
  - The user interface will be built using HTML, CSS, and JavaScript.
- **Back-End:**
  - Firebase will be used to handle back-end operations and manage the database.
  - Firestore, Firebase's NoSQL database, will be utilized for storing and retrieving data.
  - Firebase Authentication will manage user login and registration processes.

Hardware Requirements:

- **Fast Personal Computer:**
  - A high-performance personal computer is necessary for the development, coding, and testing processes.
  - It must offer adequate processing power and memory to support the development needs effectively.
- **Version Control System (Git):**
  - Git is essential for managing versions of the code and facilitating collaborative development.
  - It enables efficient teamwork by tracking code changes and maintaining version control.

2.5 Project Timeline and Schedule

Table 2.2: Project Phases and Duration

Phases	**Duration**
Requirement Analysis	2 weeks
Design and Planning	2 weeks
Web Development	5 weeks
Testing and Quality Assurance	2 weeks
Deployment	1 week
Maintenance and Support (Ongoing)	- (Continuous)

2.6 Social, Cultural, and Environmental Impact of the Research Portal Website

- **Social Impact:** The research portal will notably improve access to academic resources, and create a more connected academic community. By offering a unified platform for undergraduate thesis projects, it will make collaboration between students and teachers easier. This will not only increase academic engagement, but also allow students to participate to ongoing research and discussions in their subjects.
- **Cultural Impact:** The portal will play a key role in shaping the university's academic culture by encouraging a focus on research-driven learning and collaboration. It will highlight a wide range of topics and research efforts.Furthermore, it will encourage digital literacy and creativity by integrating technology into the daily academic experience.
- **Environmental Impact:** The shift to a digital platform will reduce the need for paper-based resources, contributing to the reduction of paper waste. This initiative supports sustainability efforts by minimizing the environmental impact associated with printing, storing, and distributing research documents.

Requirement Specifications

3.1 Stakeholders for the System

The stakeholders for the CUET Thesis & Project Portal include individuals and groups who are directly or indirectly involved with the system. These stakeholders are categorized based on their roles and interactions with the portal.

Primary Stakeholders

These stakeholders directly interact with the system and are its primary users.

- **Students:**
  - Access the portal to browse, view, and download undergraduate thesis papers.
  - Provide feedback on the usefulness of the papers.
  - Manage their profile information and passwords.
- **Faculty Members**
  - **As General Faculty:**
    - Receive notifications when a student submits a project under their course.
    - Provide feedback on any thesis paper or course project submitted to the portal.
    - Recommend outstanding papers or projects for publication in journals, conferences, or internal/external recognition.
  - **As Supervisors:**
    - Receive notifications when a student submits a thesis under their supervision.
    - Log in to review and approve/reject thesis submissions.
    - Provide feedback on supervised papers/projects.

Secondary Stakeholders

These stakeholders ensure the system's smooth operation and maintenance.

- **IICT (Institute of Information and Communication Technology):**
  - Provide and manage student login credentials.
  - Ensure system security, user authentication, and data management.
  - Offer technical support and system maintenance.
- **Developers:**
  - Design, develop, and maintain the CUET Thesis & Project Portal.
  - Ensure the portal meets functional and non-functional requirements.
  - Address bugs, implement new features, and ensure system scalability and performance.
  - Collaborate with IICT to ensure smooth integration and operation.

3.2 Software Requirements Specification (SRS) for Data Analysis System

3.2.1 Introduction

Purpose

The purpose of this document is to specify the requirements for the data analysis system of the CUET Thesis & Research Portal. This system processes uploaded papers and projects to generate visual insights, statistics, and keyword trends to help students and teachers analyze research contributions effectively.

Scope

The data analysis system analyzes uploaded data to produce research statistics, supervisor metrics, and keyword trends. It helps users explore research patterns, identify popular topics, and view supervisor contributions using interactive graphs and charts.

3.2.2 Overall Description

Product Perspective

The data analysis system includes the following key functionalities:

- **Research Statistics:** Displays aggregate counts and trends for submitted papers and projects.
- **Supervisor Analysis:** Tracks contributions of supervisors based on the number of papers supervised.
- **Keyword Trends:** Highlights popular research topics by analyzing keywords.
- **Data Visualization:** Provides charts, graphs, and interactive visuals for easy data interpretation.

User Classes and Characteristics

- **Students:**
  - View research trends and keyword analysis for project ideas.
  - Analyze supervisor contributions and related research areas.
- **Teachers:**
  - Monitor their supervised papers and identify research trends.
  - View statistics related to overall thesis submissions and topics.

3.2.3 Operating Environment

- **Platform:** Web-based, accessible via standard browsers.
- **Server:** Centralized server processes data and analytics.
- **Database:** Stores paper details, keywords, and supervisor data securely.
- **Visualization Tools:** Data rendering tools like Chart.js or D3.js.

3.2.4 Functional Requirements

Research Statistics

- **Input:** Paper and project data uploaded by students.
- **Process:** Aggregate and calculate research-related statistics.
- **Output:** Display total paper counts, project categories, and research trends.

Supervisor Analysis

- **Input:** Supervisor details linked to submitted papers.
- **Process:** Analyze and generate supervisor-based metrics.
- **Output:** Visualize supervisor contributions, such as papers supervised.

Keyword Trends

- **Input:** Keywords extracted from paper titles or abstracts.
- **Process:** Analyze keyword frequency and trends.
- **Output:** Generate keyword clouds or trend graphs showing popular topics.

Data Visualization

- **Input:** Analyzed research and supervisor data.
- **Process:** Convert raw statistics into interactive charts and graphs.
- **Output:** Line graphs, bar charts, and pie charts for visual interpretation.

3.2.5 Non-Functional Requirements

Performance

- The system must generate visual outputs within 3-5 seconds for typical datasets.

Usability

- Visualizations must be intuitive, clear, and easy to interpret for all users.

Scalability

- The system should handle increasing data volumes without performance degradation.

Security

- All data operations must comply with user access permissions.
- Visualization and analysis outputs must respect user privacy.

3.2.6 Constraints

- Data inputs must follow consistent formatting for accurate analysis.
- Visualization libraries must be compatible with the hosting environment.

3.2.7 Assumptions and Dependencies

Assumptions

- Uploaded data (papers, keywords) is accurate and properly formatted.
- The database will provide consistent and up-to-date data.

Dependencies

- The system relies on database integration for research data.
- Data visualization depends on libraries like Chart.js or D3.js.

3.2.8 Future Enhancements

- **Predictive Analytics:** Forecast future research trends based on historical data.
- **Custom Dashboards:** Allow users to filter and personalize analytical reports.
- **Export Reports:** Add functionality to download reports as PDF or CSV files.

3.2.9 Conclusion

The data analysis system in the CUET Thesis & Research Portal offers powerful tools for analyzing research data, tracking supervisor contributions, and identifying keyword

trends. By delivering clear, interactive visual insights, it enhances research understanding and supports academic exploration.

3.3 Software Requirements Specification (SRS) for Thesis & Project Management Module

3.3.1 Introduction

Purpose

The purpose of this document is to define the requirements for the Thesis & Project Management Module in the CUET Thesis & Project Portal. This module focuses on facilitating the management and interaction between students and teachers in managing thesis papers and projects. Core functionalities include viewing, downloading, recommending, providing feedback, and searching for submissions, ensuring a streamlined and collaborative academic environment.

Scope

The Thesis & Project Management Module is a vital component of the CUET Thesis & Project Portal, designed to allow students and teachers to effectively manage academic submissions. Students can upload, view, and download their thesis papers or projects, while teachers can filter, search, and provide recommendations. This module integrates with the server to ensure secure and efficient processing of user requests.

3.3.2 Overall Description

Product Perspective

The Thesis & Project Management Module is part of the broader CUET Thesis & Project Portal, interacting closely with other modules like profile management, submission tracking, and feedback systems. It utilizes a centralized server for data storage and retrieval, ensuring scalability and efficient handling of user requests.

User Classes and Characteristics

- **Students:**
  - Can view and download thesis papers or projects.
  - Provide feedback on viewed submissions.
- **Teachers:**
  - Can view, download, and filter thesis papers or projects.
  - Provide recommendations for outstanding submissions.
  - View supervisor details related to submissions.

3.3.3 Operating Environment

- **Platform:** The module is web-based, requiring standard browsers like Chrome or Firefox.
- **Server:** All requests interact with a centralized server for secure data processing.
- **Database:** A relational database system supports storage and retrieval of user submissions.

3.3.4 Functional Requirements

View Submissions

- **View Paper:** Students and teachers can view thesis papers stored in the system.
- **View Project:** Users can view course projects uploaded by students.

Feedback Management

- **Give Feedback:** Students and teachers can extend feedback on thesis papers and projects viewed.

Download Submissions

- **Download Paper/Project:** Users can download thesis papers or projects for offline use.

Search and Filter

- **Filter and Search:** Teachers can search submissions based on keywords, supervisors, or submission status.

Supervisor Details

- **View Supervisors:** Teachers can view details of the supervisors associated with the thesis or projects.

Recommendations

- **Recommend Paper/Project:** Teachers can recommend specific papers or projects for journal submissions or academic conferences.

3.3.5 Non-Functional Requirements

Performance

- The system should process user requests (view/download/recommend) in under 3 seconds.
- Support up to 100 simultaneous users without performance degradation.

Usability

- The module's interface should be intuitive and easy to navigate for both students and teachers.
- Feedback functionalities should provide clear prompts and confirmations.

Security

- User interactions with the server must be encrypted using SSL/TLS.
- Role-based access control should restrict access to sensitive data.

3.3.6 Constraints

- The system must support only PDF submissions for projects and thesis papers.
- It must adhere to web accessibility standards to ensure usability by all users.
- It must integrate seamlessly with the research portal's centralized server.

3.3.7 Assumptions and Dependencies

Assumptions

- Students will submit properly formatted documents as per the system guidelines.
- Teachers will perform timely reviews and recommendations for submissions.

Dependencies

- The module relies on the availability of the centralized server for data handling.
- It depends on a stable and secure database connection for storing and retrieving submissions.

3.3.8 Future Enhancements

- **Real-Time Notifications:** Notify users about submission feedback or recommendations in real-time.
- **AI-Based Feedback:** Introduce AI-based suggestions for improving thesis papers or projects based on uploaded content.

3.3.9 Conclusion

The Thesis & Project Management Module of the CUET Thesis & Project Portal aims to facilitate streamlined management of academic submissions. By enabling functionalities such as viewing, downloading, feedback, and recommendations, the module enhances collaboration and efficiency for students and teachers. Future enhancements, such as AI-based feedback and real-time notifications, promise to elevate its utility further. The successful implementation of this module will contribute to a robust and efficient academic research ecosystem.

3.4 Software Requirements Specification (SRS) for Profile Management Module

3.4.1 Introduction

Purpose

The purpose of this document is to outline the requirements for the Profile Management Module in the CUET Thesis & Project Portal. This module will handle user profile management, including viewing, updating, and managing actions such as uploading papers/projects, reviewing submissions, and tracking statuses. It serves as a vital component of the portal, ensuring that users can seamlessly interact with the system to fulfill their academic and administrative needs.

Scope

The Profile Management Module is a key part of the CUET Thesis & Project Portal, designed to enable effective management of user profiles and their activities. It allows students to upload thesis papers or course projects, specify supervisors, and track the review process. Teachers can review and accept/reject these submissions, as well as provide feedback. This module integrates with other functionalities of the research portal to streamline the workflow and improve user experience.

3.4.2 Overall Description

Product Perspective

The Profile Management Module is tightly integrated within the CUET Thesis & Project Portal. It interacts with modules responsible for submission management and feedback tracking. It leverages a centralized database for storing user profiles and submission records.

User Classes and Characteristics

- **Students:** Students can view and update their profiles, upload thesis papers or projects, and track submission statuses.
- **Teachers:** Teachers can view and update their profiles, review thesis papers or projects, accept/reject submissions, and provide feedback.

3.4.3 Operating Environment

The module is web-based and accessible via standard web browsers. It relies on a reliable database for storing user data and submission details and supports secure access for authorized users.

3.4.4 Functional Requirements

Profile Management

- **View Profile:** The system shall allow users to view their profile information, such as name, department, and account details.
- **Update Profile:** The system shall enable users to update their profile details and validate the input before saving changes.

Submission Management

- **Upload Paper/Project:** Students shall be able to upload their thesis papers and course projects in PDF format. The system shall require them to specify their supervisor while uploading.
- **View Submission Status:** Students shall be able to view the status of their submissions, categorized as "Pending Review," "Accepted," or "Rejected."

Review Process

- **Review Thesis and Projects:** Teachers shall be able to review thesis papers and course projects. They can leave feedback for students during this process.
- **Accept/Reject Submissions:** The system shall provide teachers with an option to accept or reject submissions based on their review.

Search History

- **View Search History:** Students shall be able to view their search history for previously accessed materials in the portal.

3.4.5 Non-Functional Requirements

Performance

- The system shall retrieve and update profile information within 3 seconds.
- It shall support simultaneous updates by multiple users without conflicts.

Usability

- The profile interface shall be intuitive and user-friendly.
- Error messages shall be clear and provide guidance on corrective actions.

Security

- User profile data shall be protected through encryption and role-based access control.
- Only authorized users shall have access to sensitive functionalities.

3.4.6 Constraints

- The system must adhere to web accessibility standards to ensure usability for diverse users.
- The module must integrate seamlessly with the research portal's database and existing modules.

3.4.7 Assumptions and Dependencies

Assumptions

- The system assumes that all users (students and teachers) will have access to the internet and a compatible web browser to use the platform.
- It is assumed that all submission data provided by the students will be accurate and correctly formatted.
- The system assumes that teachers and supervisors will review submissions promptly.

Dependencies

- The system depends on the availability and stability of the centralized database for storing and retrieving user profiles and submissions.
- It relies on secure login mechanisms for students and teachers to ensure data privacy.
- The module is dependent on existing modules for feedback tracking and publishing records.

3.4.8 Future Enhancements

- **Advanced Notification System:** The system could be enhanced to include a notification system for both students and teachers, such as email or in-app notifications for updates on review status or profile changes.
- **Analytics Dashboard:** An analytics dashboard could be developed for teachers to track the number of papers/projects reviewed and for students to view statistics on their uploaded submissions.
- **Integration with Publication Platforms:** Future versions of the module could be integrated with external publication platforms to allow automatic linking of accepted papers/projects to external journals or conference websites.

3.4.9 Conclusion

The Profile Management Module of the CUET Thesis & Project Portal aims to facilitate seamless interaction between students and teachers for managing academic profiles and submissions. By enabling features such as profile updating, paper/project uploads, and review management, the module ensures efficient collaboration. Future enhancements, such as advanced notifications and analytics, will further improve its functionality. The successful implementation of this module will significantly contribute to creating a well-structured academic research ecosystem.

3.5 Use case diagram with Graphical and Textual Description

3.5.1 Use Case Diagram of Data Analysis

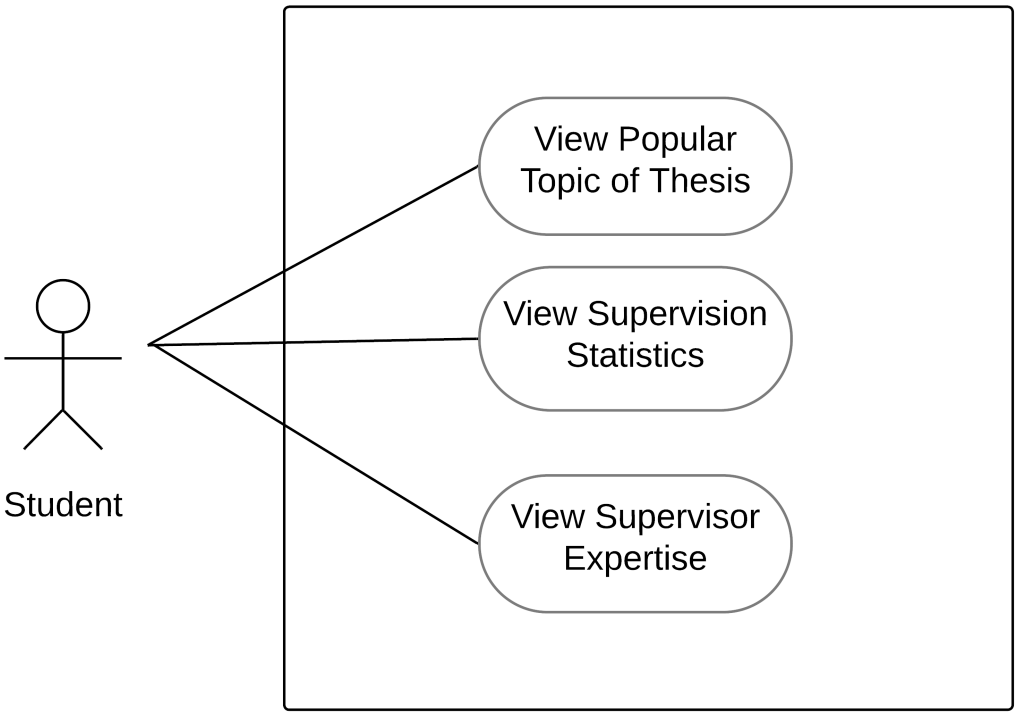


Figure 3.1: Use Case Diagram of Data Analysis

Use Case: View Thesis Insights (Popular Topics, Supervision Statistics, and Supervisor Expertise) **Iteration:** 3
Primary Actor: Admin **Goal in Context:** To provide administrators with comprehensive insights into thesis-related data, including popular topics, supervision statistics, and supervisor expertise. **Preconditions:**

- The admin must be logged into the system.
- Thesis and supervision data must exist in the database.

Trigger: The admin decides to analyze thesis trends, supervisor contributions, and expertise. **Scenario:**

1. The admin logs into the system.
2. The admin selects the "Thesis Insights" dashboard option.
3. The system fetches and processes thesis and supervision data from the database.
4. The system identifies:
    - The most popular thesis topics.
    - The number of theses supervised by each supervisor.
    - Supervisors' expertise based on thesis topics they supervised.
5. The system generates and displays results in charts, graphs, or lists for better visualization.

Exceptions:

- If no thesis or supervision data exists, an appropriate message is displayed.
- If there is a system or database error, the system displays an error message.

Priority: High priority, as part of core functionality for data analysis. **When Available:** Second increment **Frequency of Use:** Frequent **Channel to Actor:** Via a web browser on a PC or mobile device, using an internet connection. **Secondary Actors:** Server **Channels to Secondary Actors:**

- Retrieving thesis and supervision records from the database.
- Processing and generating insights for display.

Open Issues:

- Ensuring real-time updates of insights as new data is added.
- Optimizing performance for large datasets.
- Adding filtering options (e.g., by department or time range) for deeper analysis.

3.5.2 Use Case Diagram of Thesis and Project Submission System

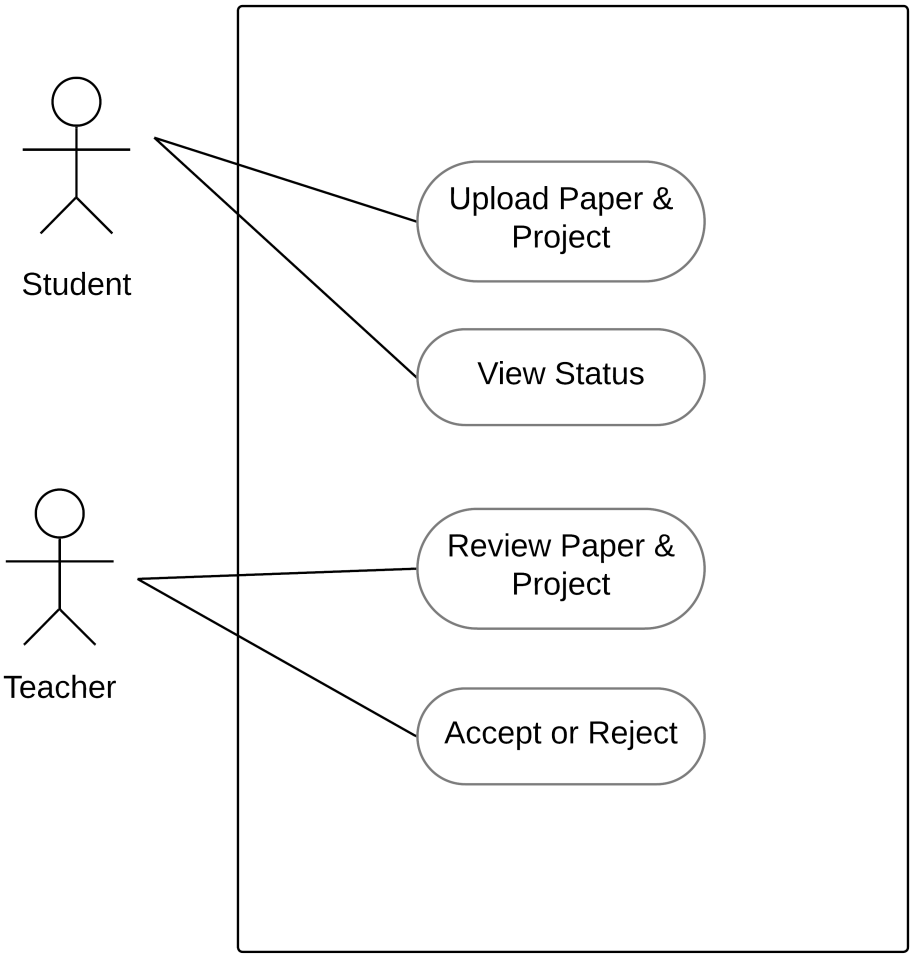


Figure 3.2: Use Case Diagram of Thesis and Project Submission System

Use Case: Thesis Paper and Project Submission System

Iteration: 1
Primary Actor: Student, Teacher **Goal in Context:** Facilitate uploading, reviewing, and managing papers and projects. **Preconditions:**

- Users must be logged into the system.
- A valid paper or project must be ready for upload.

Trigger: The user decides to upload a paper or project for review. **Scenario:**

1. The user logs into the system.
2. The user selects **Upload Paper & Project** and submits the file.
3. The reviewer logs in and selects **Review Paper & Project**.
4. The reviewer evaluates the uploaded content and performs an action.
5. The reviewer chooses **Accept or Reject** based on the evaluation.
6. The user logs in to **View Status** to check the current decision.

Exceptions:

- If file upload fails, an error message is shown.
- If no projects are available for review, the reviewer sees an empty list.

Priority: High **When Available:** First increment **Frequency of Use:** Regular **Channel to Actor:** Web application via browser **Secondary Actors:**

- Server
- Database

Channels to Secondary Actors:

- The server processes file uploads and downloads.
- The database stores and retrieves paper and project data.
- The server updates the status (accept/reject) based on reviewer actions.

Open Issues:

- Handling large file uploads efficiently.
- Adding notifications for status updates.
- Ensuring data security and preventing unauthorized access.

3.5.3 Use Case Diagram of Paper and Project Workflow

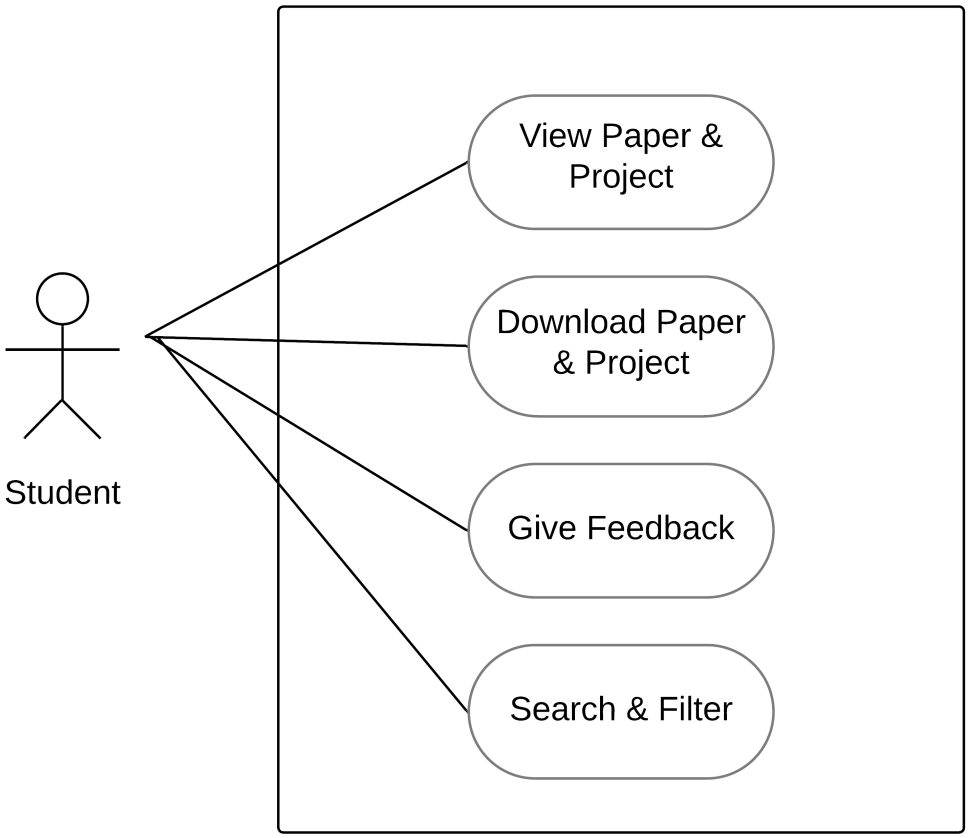


Figure 3.3: Use Case Diagram of Paper and Project Workflow

Use Case: Paper & Project Workflow **Iteration:** 2
Primary Actor: User, Reviewer **Goal in**

Context: Enable users to view, download, search, and provide feedback on papers and projects. **Preconditions:**

- The user must be logged into the system.
- Papers and projects must exist in the database.

Trigger: The user decides to interact with available papers and projects. **Scenario:**

1. The user logs into the system.
2. The user selects **View Paper & Project** to browse available submissions.
3. The user selects **Download Paper & Project** to save a copy of the chosen submission.
4. The user selects **Search & Filter** to locate specific papers or projects.
5. The user or reviewer selects **Give Feedback** to provide comments or suggestions.
6. The system processes the feedback and saves it to the database.

Exceptions:

- If no papers or projects exist, the system displays a "No Data Available" message.
- If the download fails, an error message is displayed.

Priority: High **When Available:** Second increment **Frequency of Use:** Regular **Channel to Actor:** Web application via browser **Secondary Actors:**

- Server
- Database

Channels to Secondary Actors:

- The server provides access to view and download papers and projects.
- The database stores papers, projects, and feedback provided by users.
- The system fetches filtered results when the user applies search or filter options.

Open Issues:

- Improving search efficiency for large datasets.
- Allowing offline access to downloaded files.
- Adding validation for feedback submissions to ensure content quality.

3.6 Activity Diagram

In Software Engineering, an Activity Diagram is a type of UML (Unified Modeling Language) diagram used to model the workflow of a system, showcasing the sequence of activities or actions in a process or operation. It helps software engineers and developers understand the flow of control, decision-making processes, and interactions within a system, whether it's a user action, system response, or the internal processing logic.

3.6.1 Activity Diagram of Data Analysis

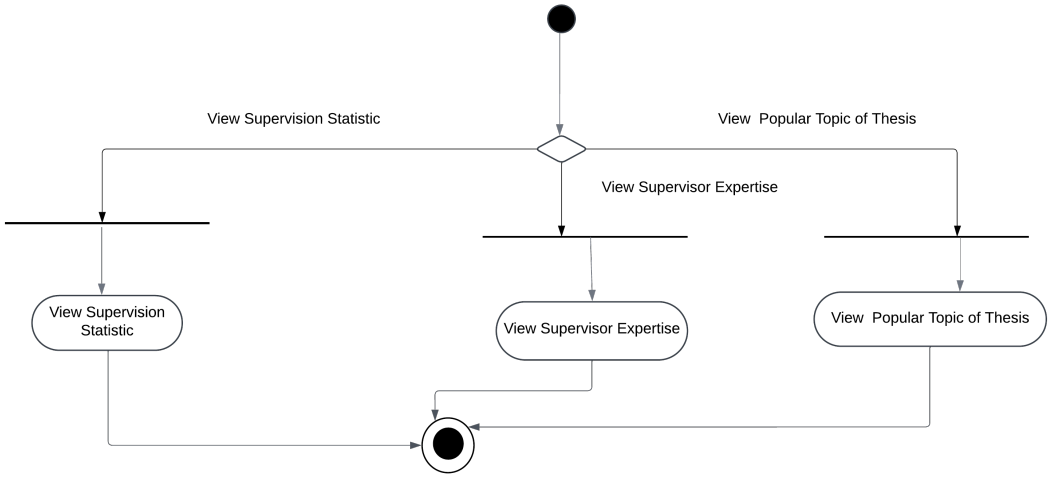


Figure 3.4: Activity Diagram of Data Analysis

3.6.2 Activity Diagram of Submission and Review Process

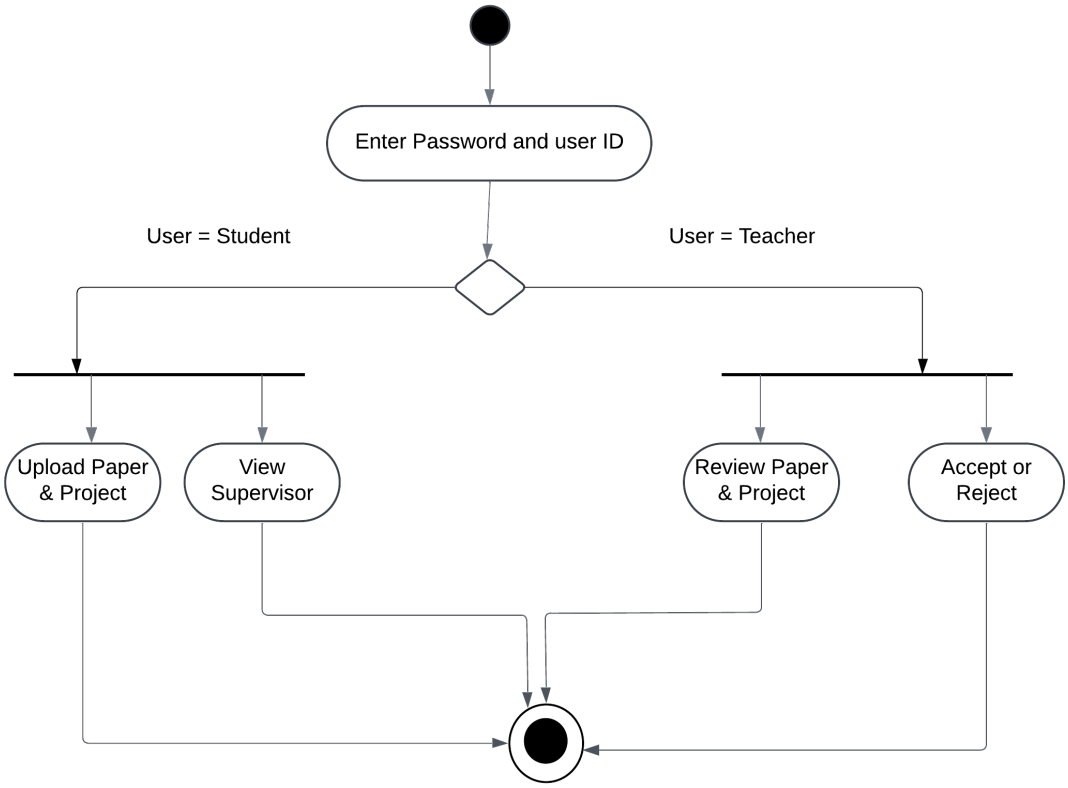


Figure 3.5: Activity Diagram of Submission and Review Process

3.6.3 Activity Diagram of Project Access and Feedback

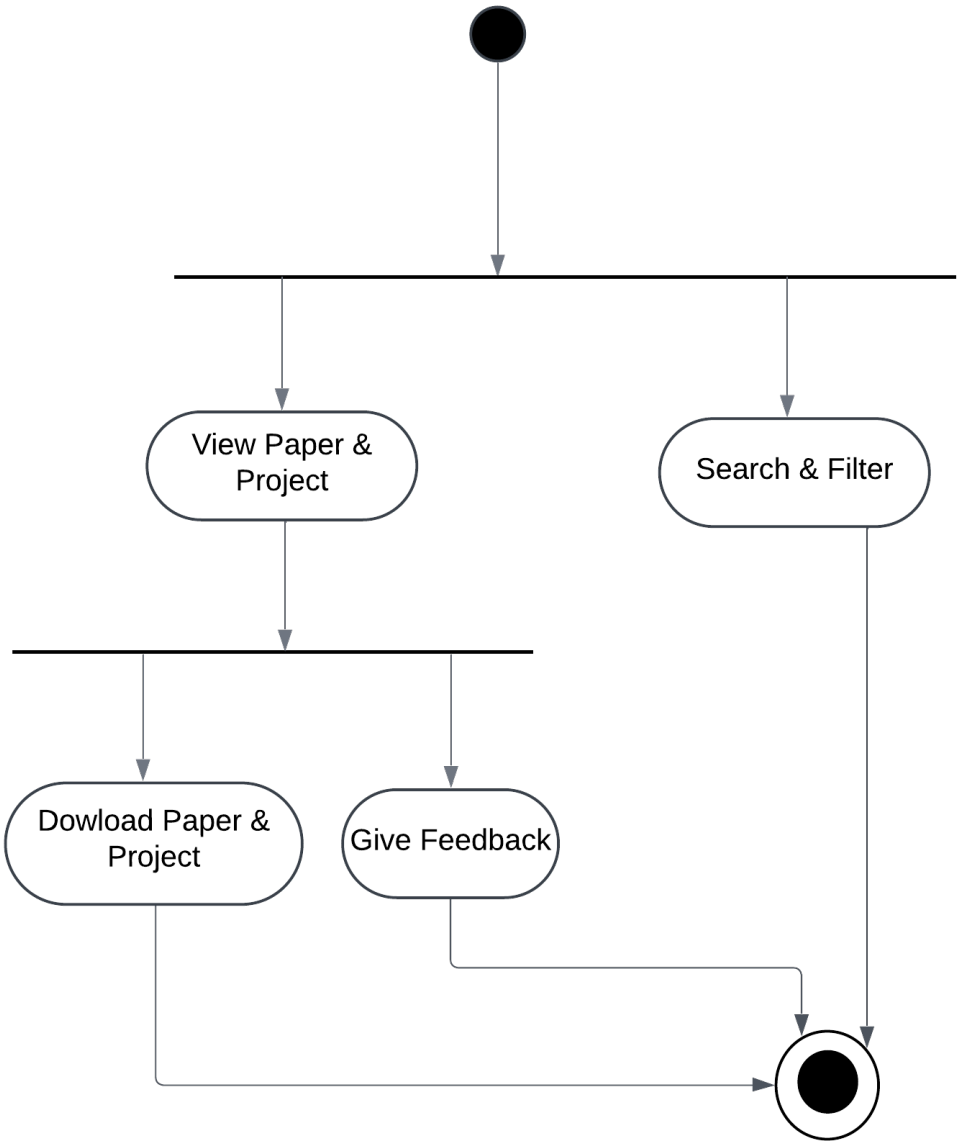


Figure 3.6: Activity Diagram of Project Search, Access and Feedback

3.7 Swimlane Diagram

A swimlane diagram is a type of flowchart that visually represents a process or workflow by organizing the steps into distinct ”lanes.” Each lane is dedicated to a specific actor, role, system, or entity responsible for carrying out those steps in the process. The lanes can be horizontal or vertical, and the purpose of a swimlane diagram is to clearly show the interactions between multiple participants or components involved in a process.

3.7.1 Swimlane Diagram of Data Analysis

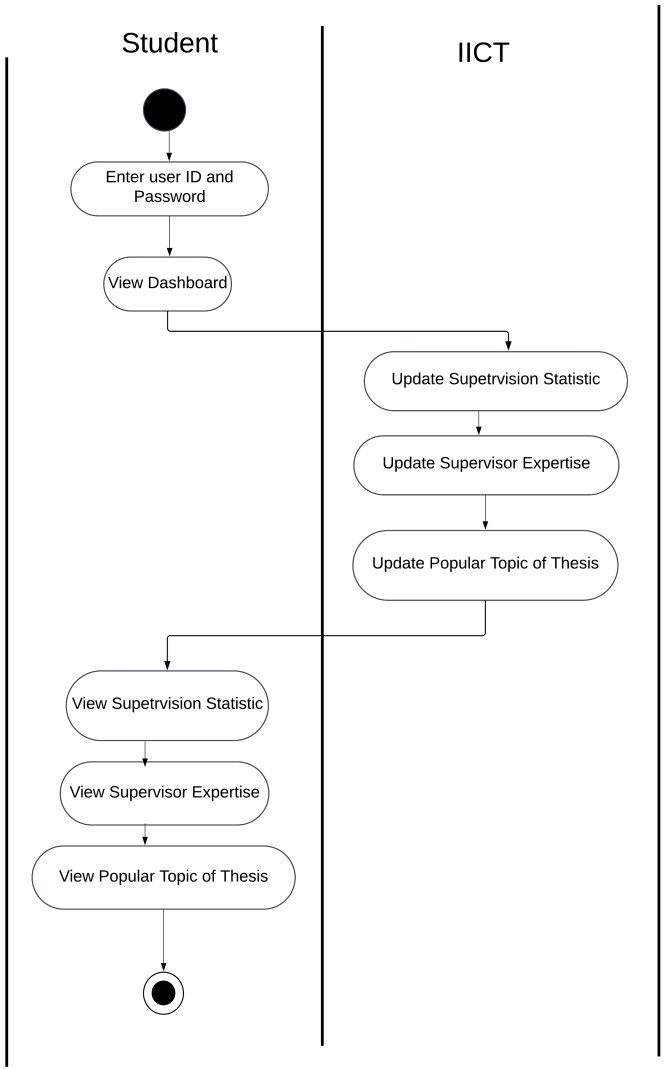


Figure 3.7: Swimlane Diagram of Data Analysis

3.7.2 Swimlane Diagram of Submission and Review Process

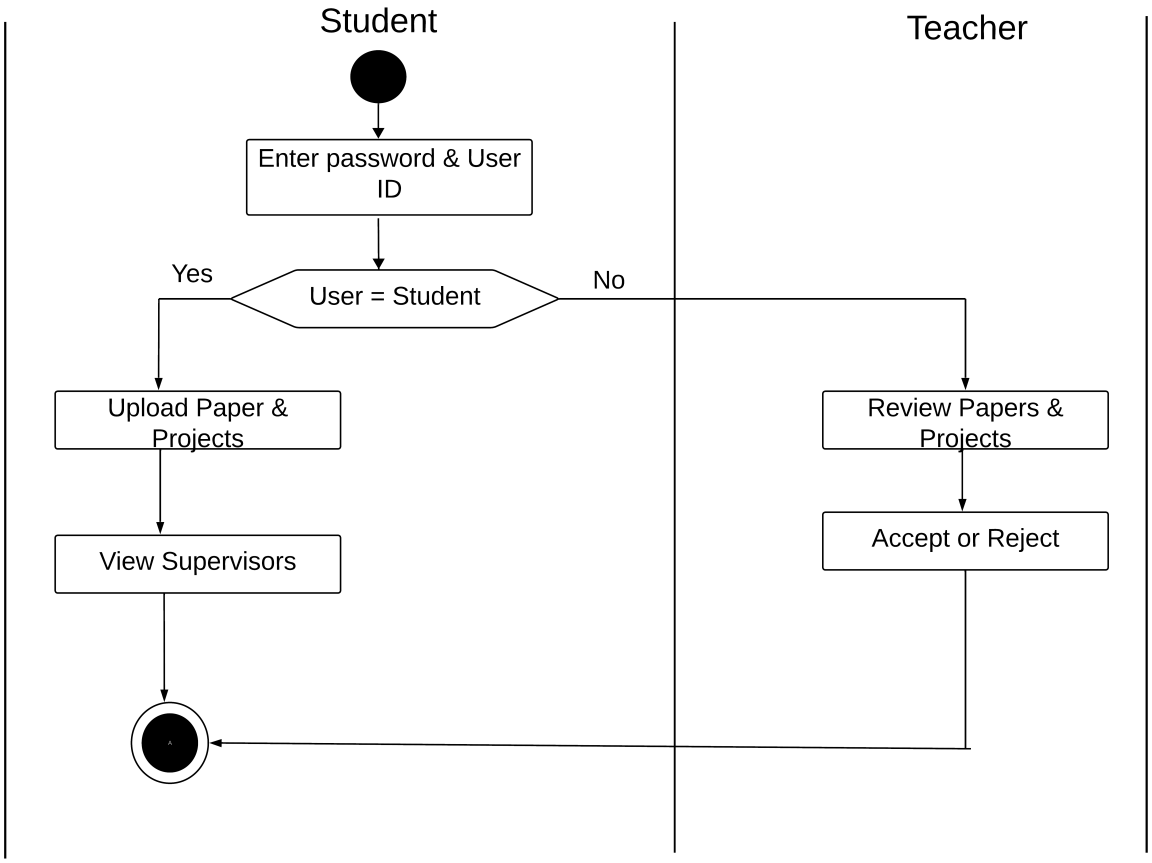


Figure 3.8: Swimlane Diagram of Submission and Review Process

3.8 Static model – class diagram

A class diagram is a static structure diagram that depicts the static view of a system. It shows the classes, their attributes, and the relationships between them. Class diagrams are used to model the structure of a system and to define the static relationships between the different classes.

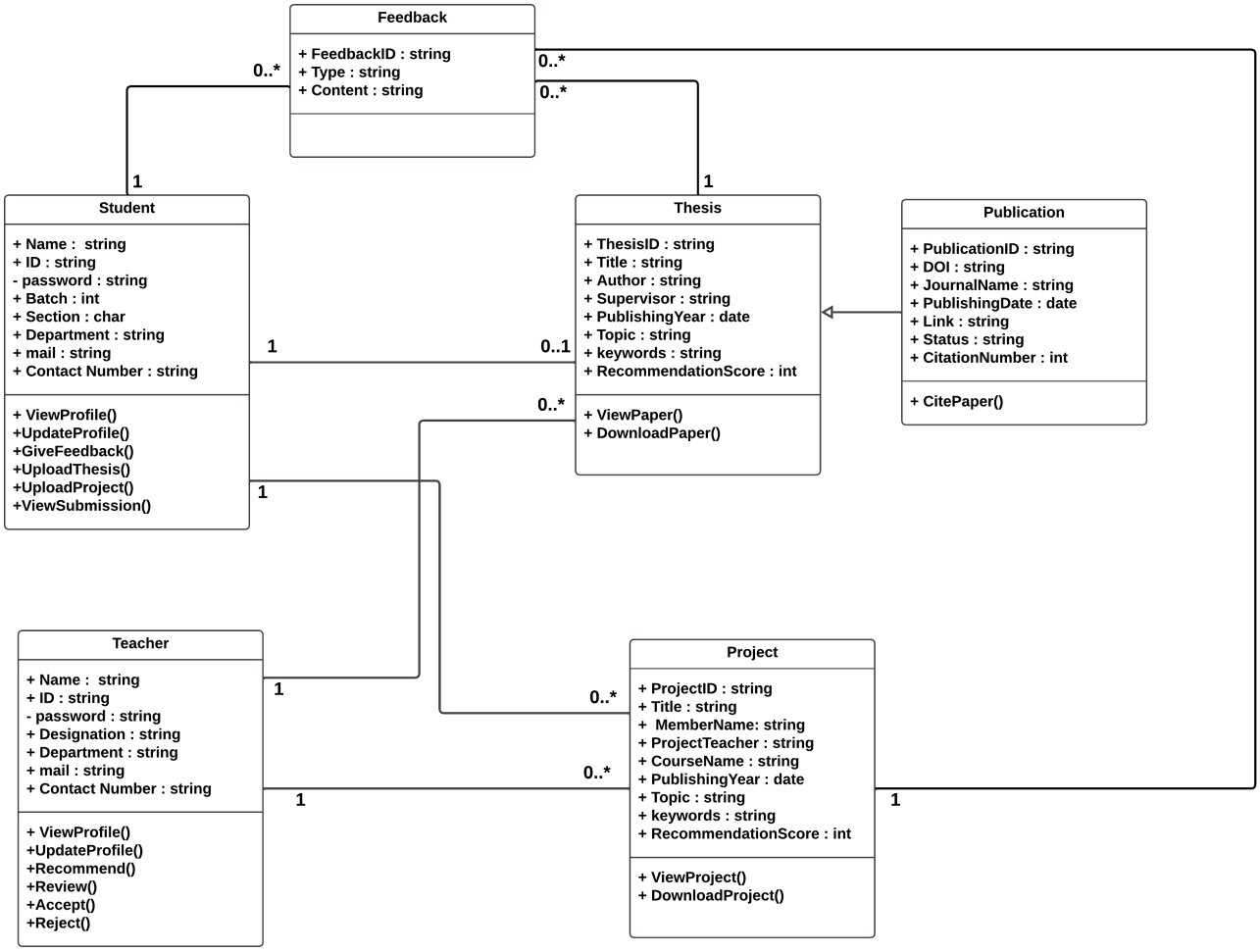


Figure 3.9: Class Diagram

3.9 Dynamic model – sequence diagram

In Software Engineering, a Sequence Diagram is a type of UML (Unified Modeling Language) diagram used to model the interaction between objects or components in a system, specifically focusing on the order and timing of messages exchanged during the execution of a use case or process. It shows how different parts of the system interact over time, emphasizing the sequence of method or function calls that occur between objects.

3.9.1 Sequence Diagram of Data Analysis

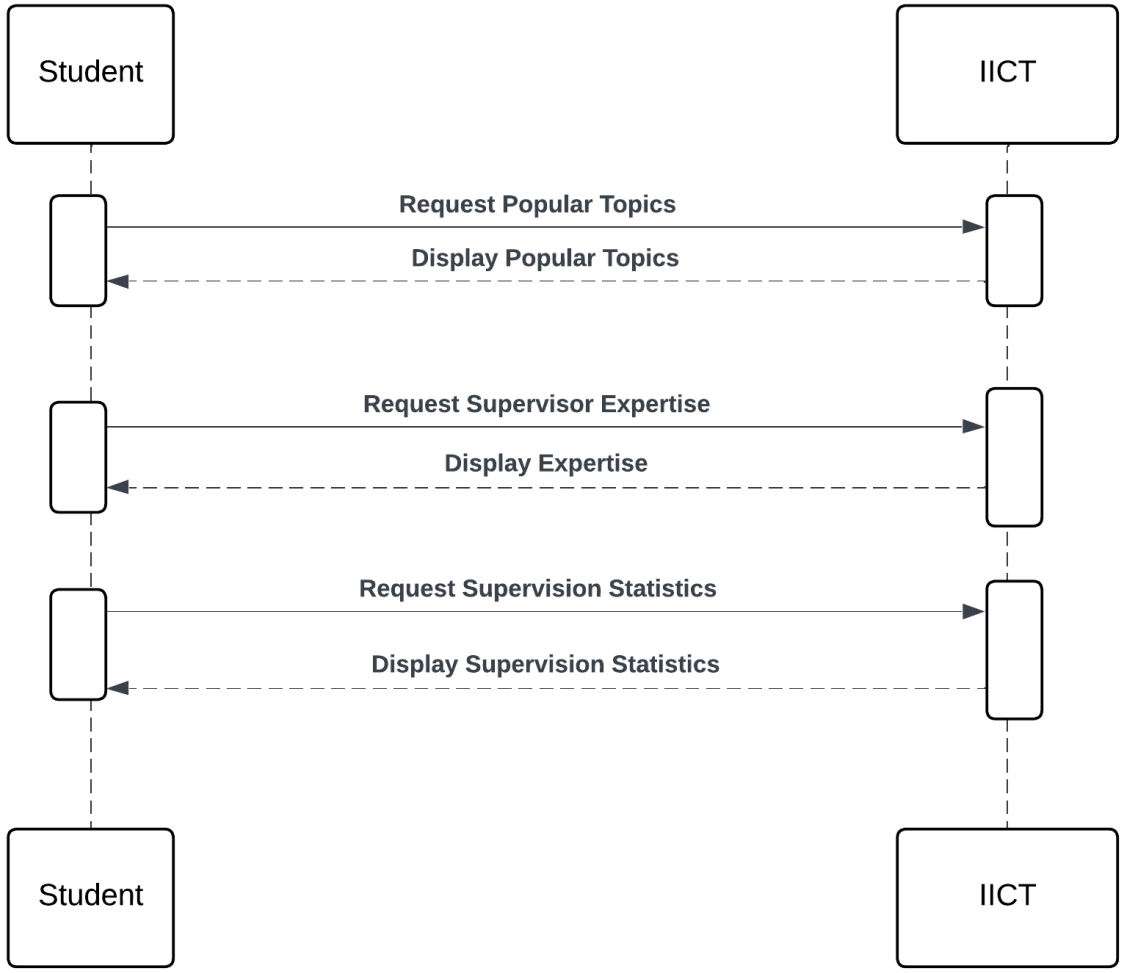


Figure 3.10: Sequence Diagram of Data Analysis

3.9.2 Sequence Diagram of Submission and Review Process

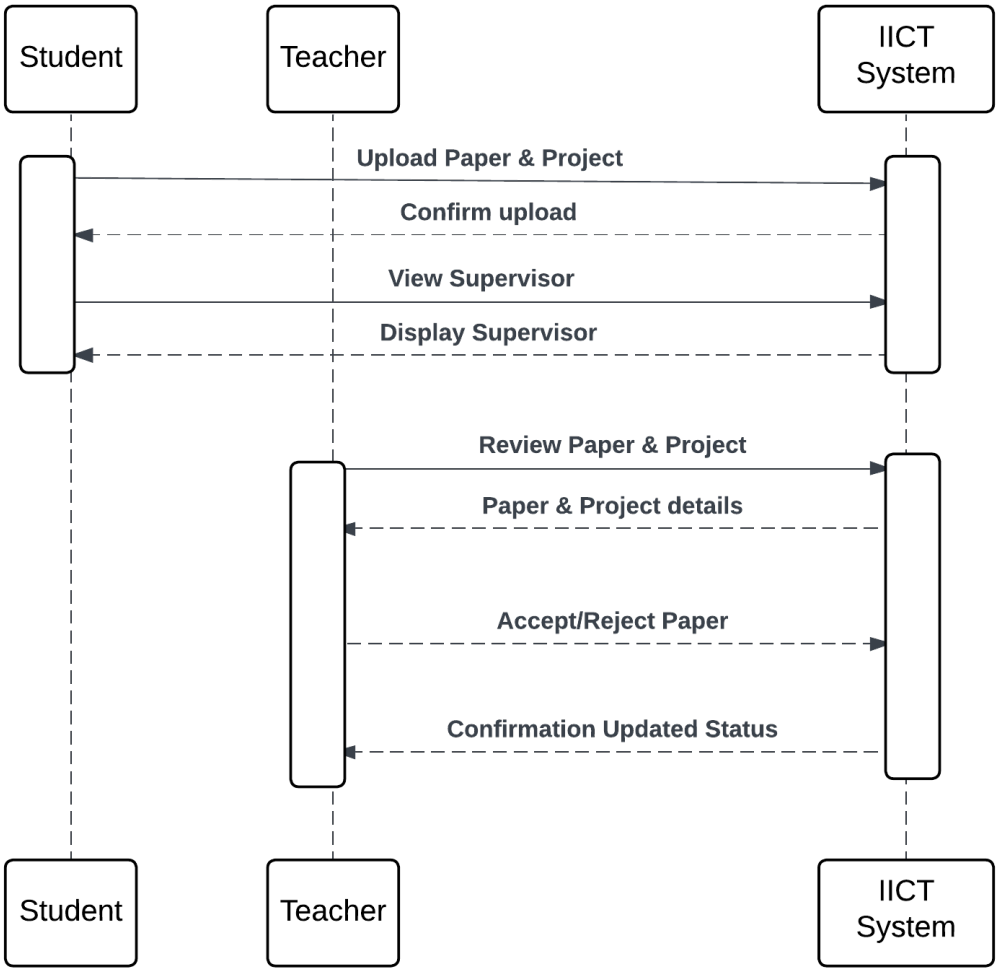


Figure 3.11: Sequence Diagram of Submission and Review Process

3.9.3 Sequence Diagram of Project Search, Access and Feedback

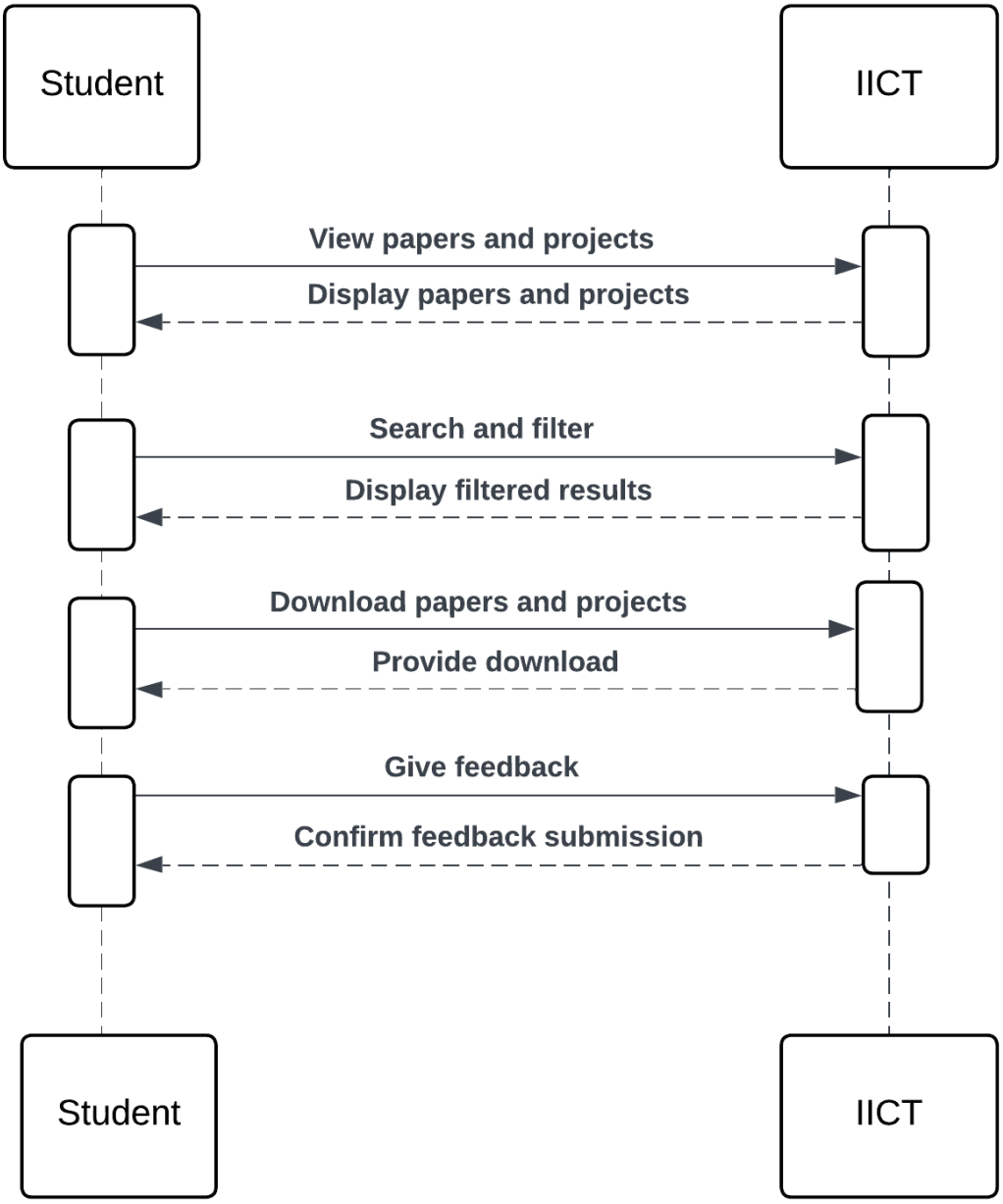


Figure 3.12: Sequence Diagram of Project Search, Access and Feedback

3.9.4 Integrity Requirements

- Data integrity checks are performed during file uploads to prevent corrupted or incomplete thesis submissions.
- Only authorized supervisors can modify the status (accepted/rejected) of a thesis.
- Version control is implemented for thesis submissions to track changes and prevent unauthorized alterations.
- Regular automated and manual database backups are scheduled to ensure data recovery in case of system failure or data loss.

3.9.5 Privacy Requirements

- All user data (e.g., student ID, email, and passwords) is encrypted in transit using SSL/TLS (HTTPS) and at rest using advanced encryption standards (AES).
- Thesis papers are accessible only to authenticated users; public access is restricted to protect sensitive academic information.
- Personal Identifiable Information (PII) is handled in compliance with applicable data protection laws and university policies.
- Users have the ability to control their profile visibility and update privacy settings.
- Access logs are maintained to monitor and audit user activity, ensuring transparency and accountability while respecting user privacy.

Architecture

4.1 Architectural model/style used

The architectural style used is the **Data Centered Architecture**.

- **Centralized Data Store:** A central repository (data store) serves as the core component, managing all data such as theses, projects, supervisor information, and profiles.
- **Client Components:** Various modules (Thesis and Project Directory, Supervisor Information, Data Analysis, Profile Information, and Searching Thesis and Project) interact with the data store for accessing or updating data.
- **Data Flow:** All components depend on the centralized data store, making it the single source of truth.

This architecture ensures *data integrity, centralized control*, and facilitates efficient data access for all components.

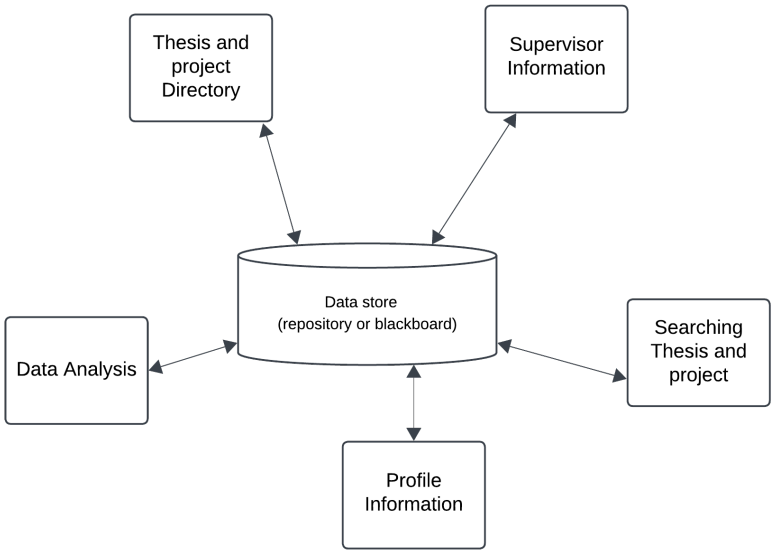


Figure 4.1: Data Centered Architecture for CUET Thesis & Project Portal

4.1.1 Rationale for choosing your architectural model/style

The selection of a **data-centered architecture** for the **CUET Thesis Portal** is driven by the need to efficiently manage and centralize various components such as thesis and project directories, supervisor information, Thesis and project searches, profile information, and data analysis. At the core of the system lies the **data store** (repository or blackboard), which serves as the central hub for storing and integrating data. The **thesis and project directory** organizes and streamlines access to works of students, while the **supervisor information** module connects students with relevant supervisors. The **searching thesis and projects** feature allows users to filter and discover relevant works of particular topics, improving accessibility and usability. Additionally, **data analysis** module derives insights and trends to enhance decision-making processes based on popular topics and thesis fields. This architectural design emphasizes centralized data management, ensuring efficient access, updates, and consistency across all modules. By centralizing the data flow, the system achieves scalability and supports functionalities such as project tracking, supervision management, and advanced analytics. Overall, the data-centered architecture provides a structured and well-organized framework, enabling seamless relationships between components and effective thesis and project management.

4.2 Technology, software, and hardware used

- **Frontend Technologies:** HTML, CSS, Bootstrap, React
  - **HTML:** Structure the web pages.
  - **CSS:** Style and enhance the user interface.
  - **Bootstrap:** Responsive design and prebuilt components.
  - **React:** Build dynamic and interactive user interfaces.
- **Backend Technologies:** Firebase
  - **Firebase Authentication:** Secure user login and authentication.
  - **Firebase Firestore:** Cloud-hosted NoSQL database for data storage.
  - **Firebase Hosting:** Deploy and host web applications.
  - **Firebase Storage:** Store and manage uploaded theses and project files.
- **Hardware:** Centralized Server
  - **Purpose:** Host the application and serve user requests efficiently.

Design

5.1 Component level design following pattern

As a higher-level interface, the component exports and wraps its implementation. It is an interchangeable, reusable, portable, and modular set of well-specified functions. A component is a segment of software that is either meant to communicate with the other parts and has a set of functions or it comprises a specific functionality. The only way to characterize the software component is as a composition unit that has other composing parts, a contractually specified interface, and obvious context dependencies.

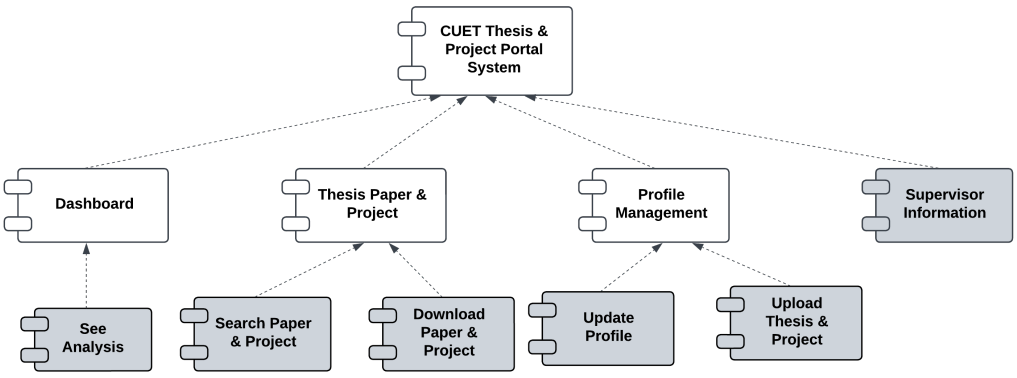


Figure 5.1: Structure Chart of CUET Thesis and Project Portal

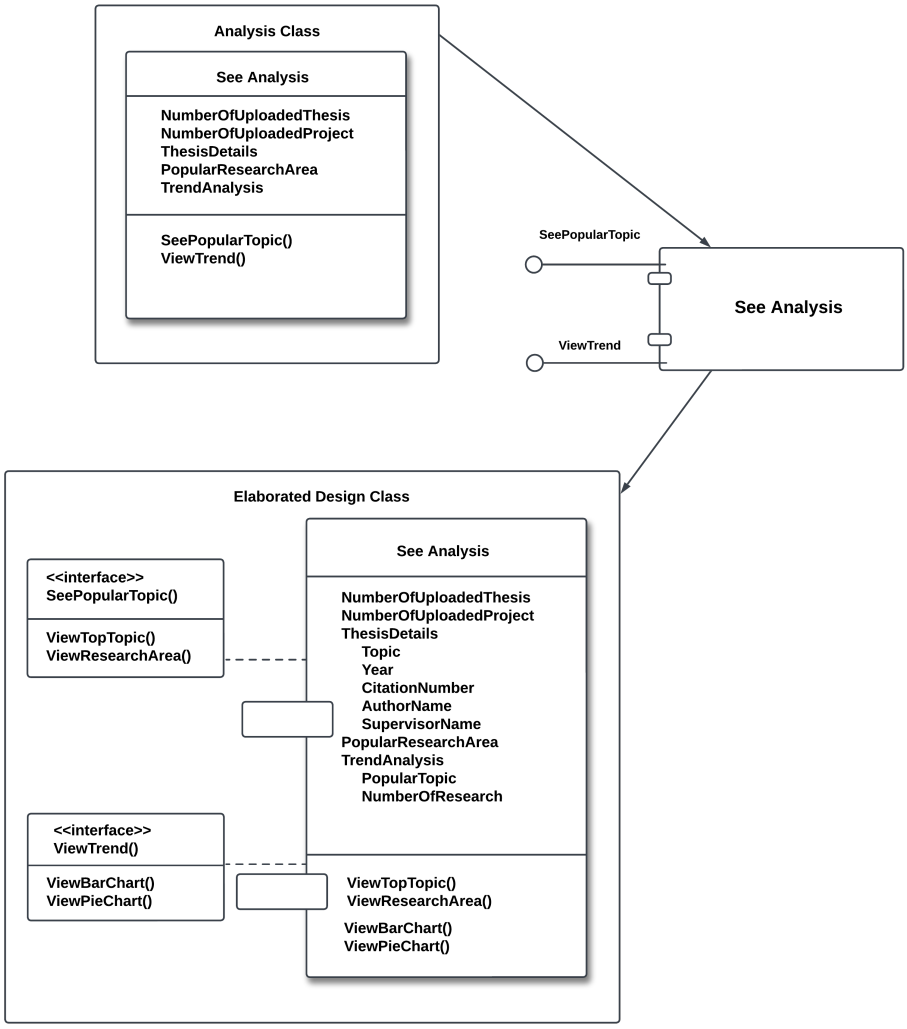


Figure 5.2: Component Level Design for See Analysis

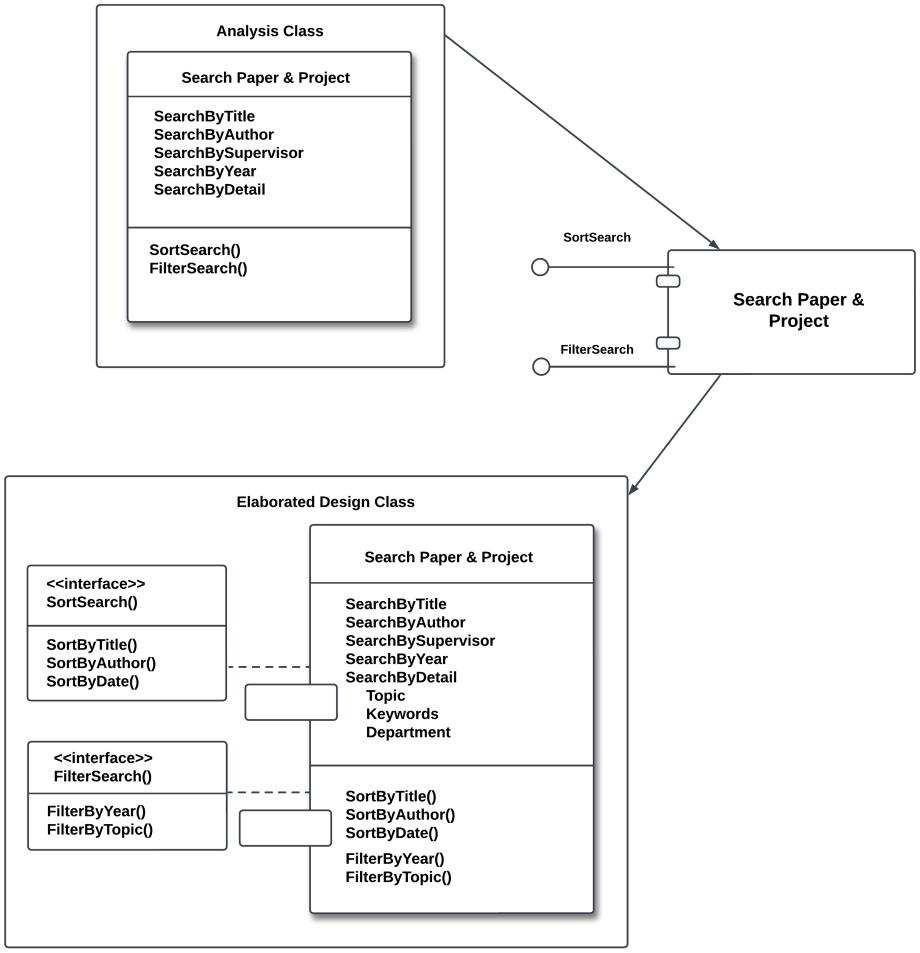


Figure 5.3: Component Level Design for Search Paper and Project

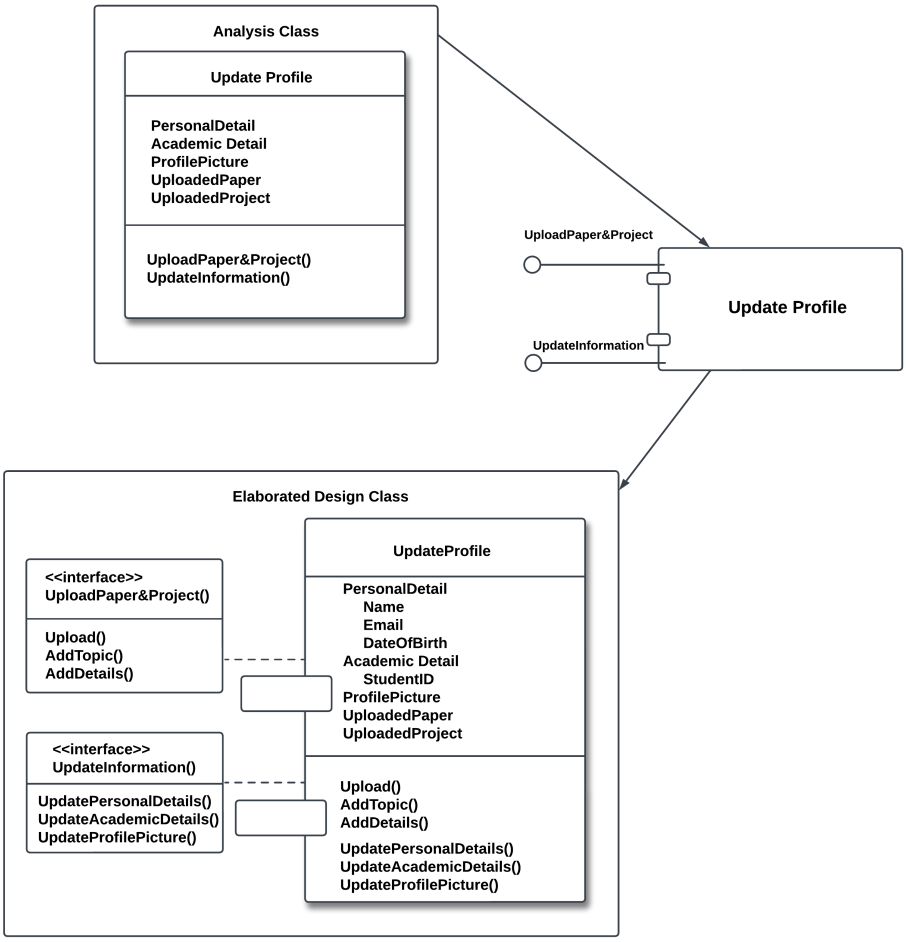


Figure 5.4: Component Level Design for Update Profile

5.2 Data Flow Diagram

The Data Flow Diagram (DFD) for the **CUET Thesis & Project Portal** illustrates the flow of data between various components within the system. The diagram captures how data is input, processed, stored, and output by different modules of the portal. It serves as a visual representation of how users interact with the system and how data is transferred between the portal's modules such as the thesis and project directory, supervisor information, profile management, and data analysis.

The DFD is structured to showcase the key processes, data stores, and data flows within the portal. It helps in understanding the relationships between various components, ensuring efficient data handling, and supporting future system enhancements. By identifying the movement of data, the DFD allows for better planning of the system architecture and contributes to an overall understanding of the system's functionality and performance.

Context Diagram

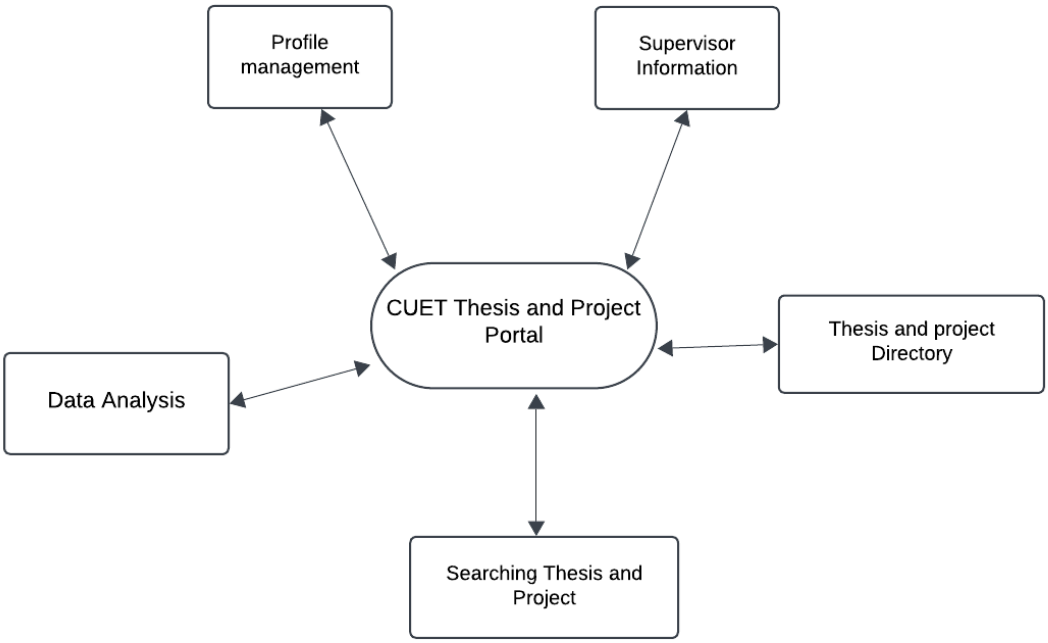


Figure 5.5: Context Diagram

Student Profile Data Flow Diagram

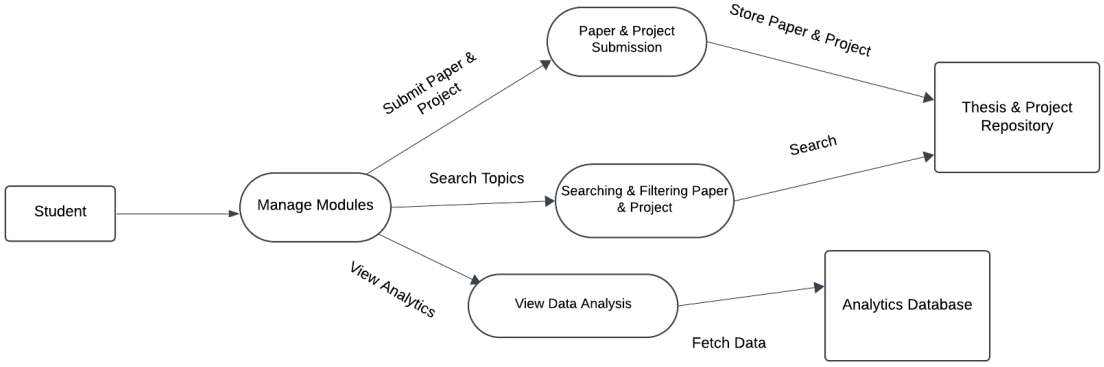


Figure 5.6: Student Profile Data Flow Diagram

Teacher Profile Data Flow Diagram

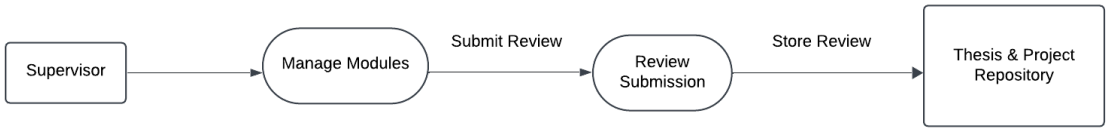


Figure 5.7: Teacher Profile Data Flow Diagram

5.3 ER Diagram

The Entity-Relationship (ER) Diagram for the **CUET Thesis & Project Portal** visually represents the key entities within the system and their relationships. It defines how entities such as students, supervisors, theses, and projects interact with one another. Each entity is depicted as a box, and the relationships between them are shown with connecting lines, often annotated with cardinality to indicate the nature of the relationships (e.g., one-to-many, many-to-many).

The ER diagram serves as a blueprint for the database design, highlighting how data is organized and related within the system. It ensures that the portal's database structure supports efficient data retrieval, integrity, and consistency. This diagram plays a crucial role in database design and helps developers understand the interdependencies between various components of the system.

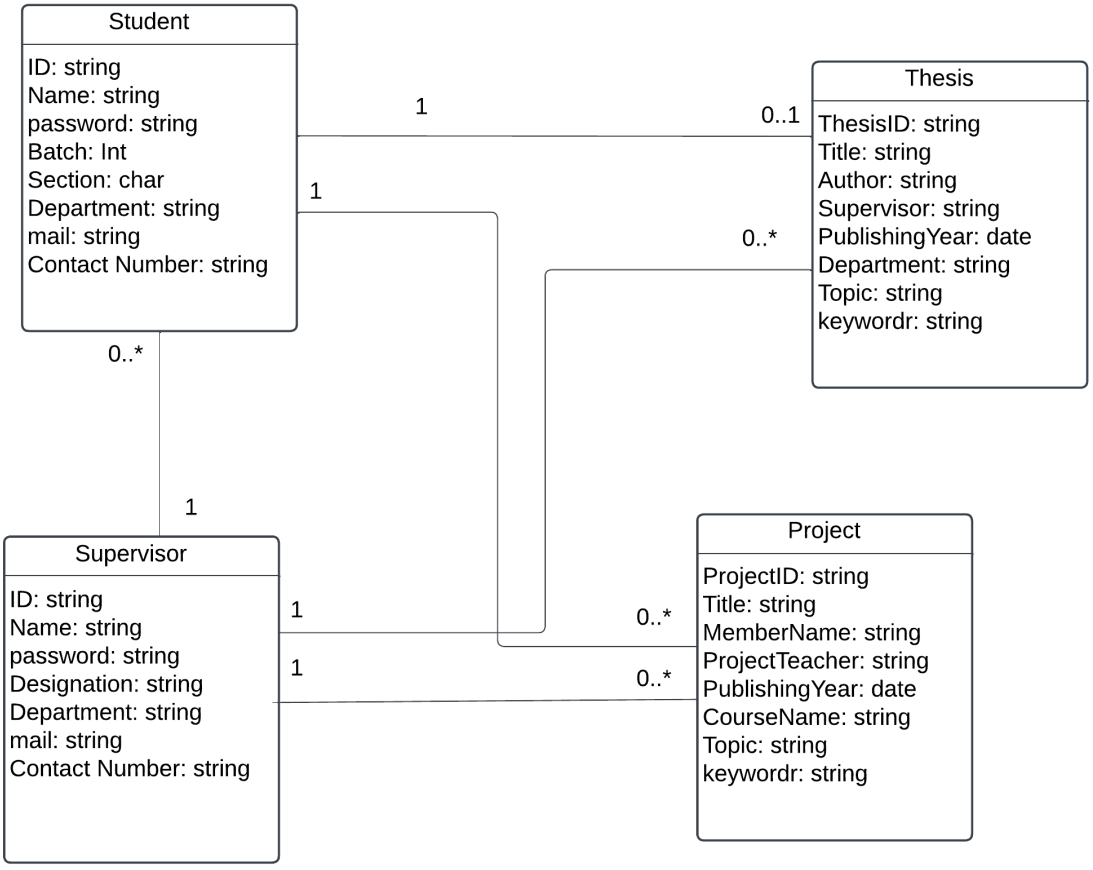


Figure 5.8: ER Diagram

5.4 UX Design

Dashboard

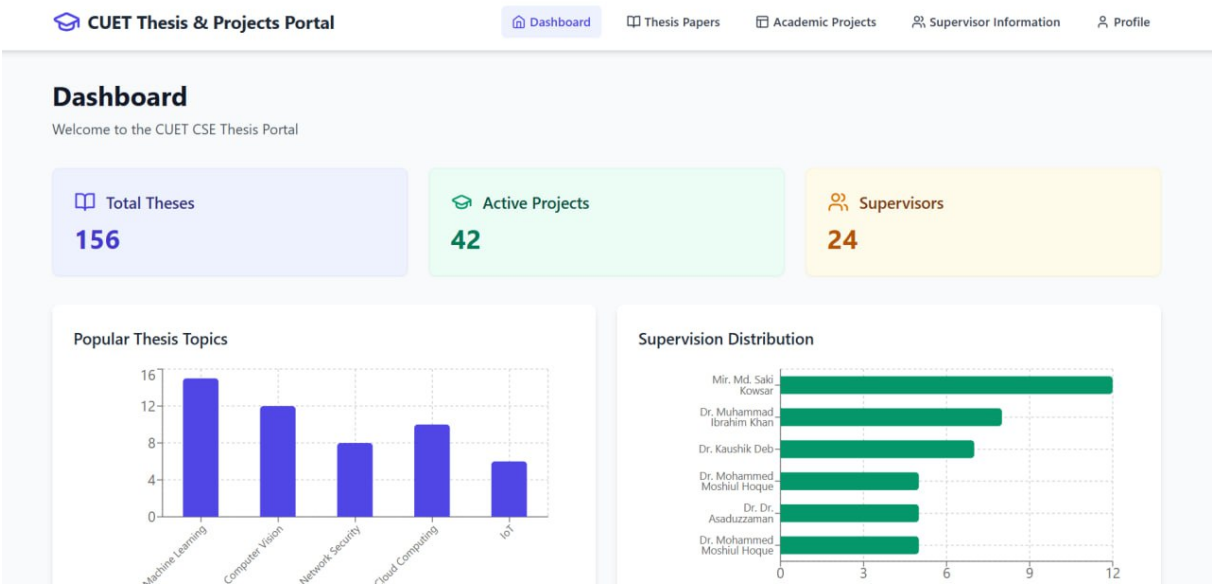


Figure 5.9: Dashboard

Thesis Paper Page

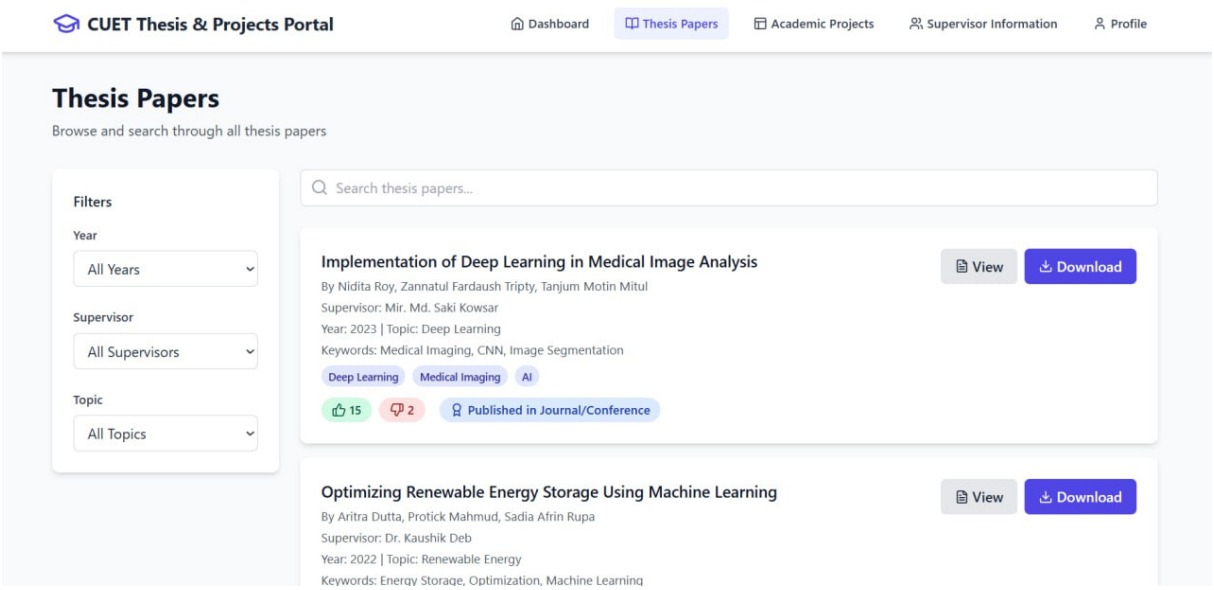


Figure 5.10: Thesis Paper Page

Publication Pop Up

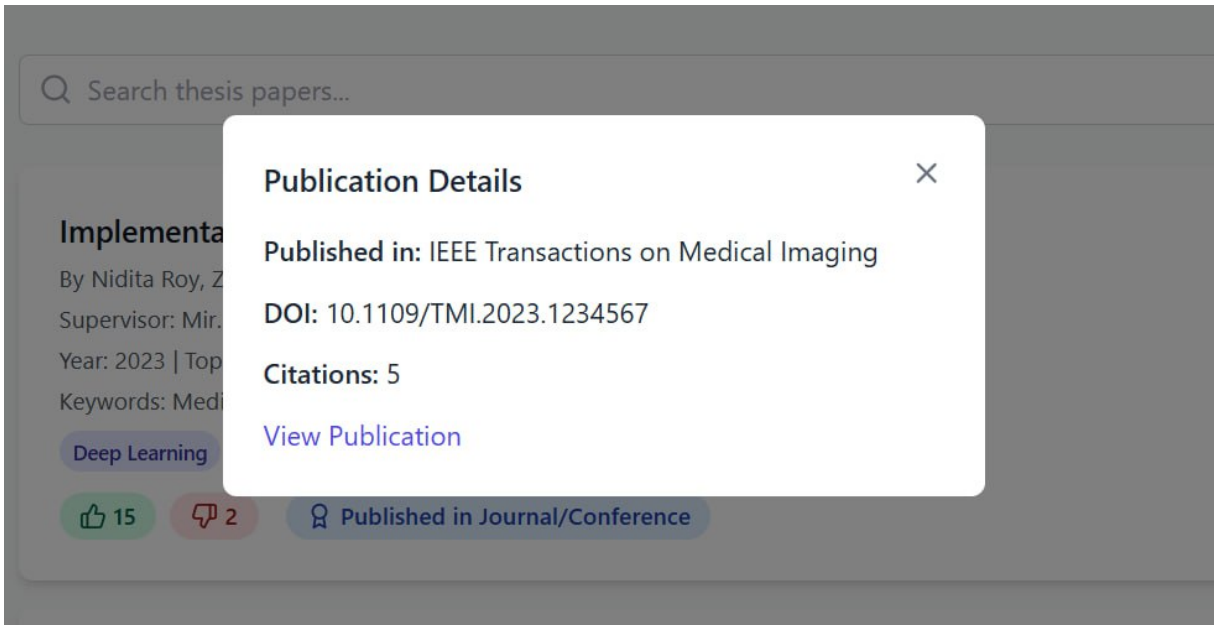


Figure 5.11: Publication pop up

Supervisor Page

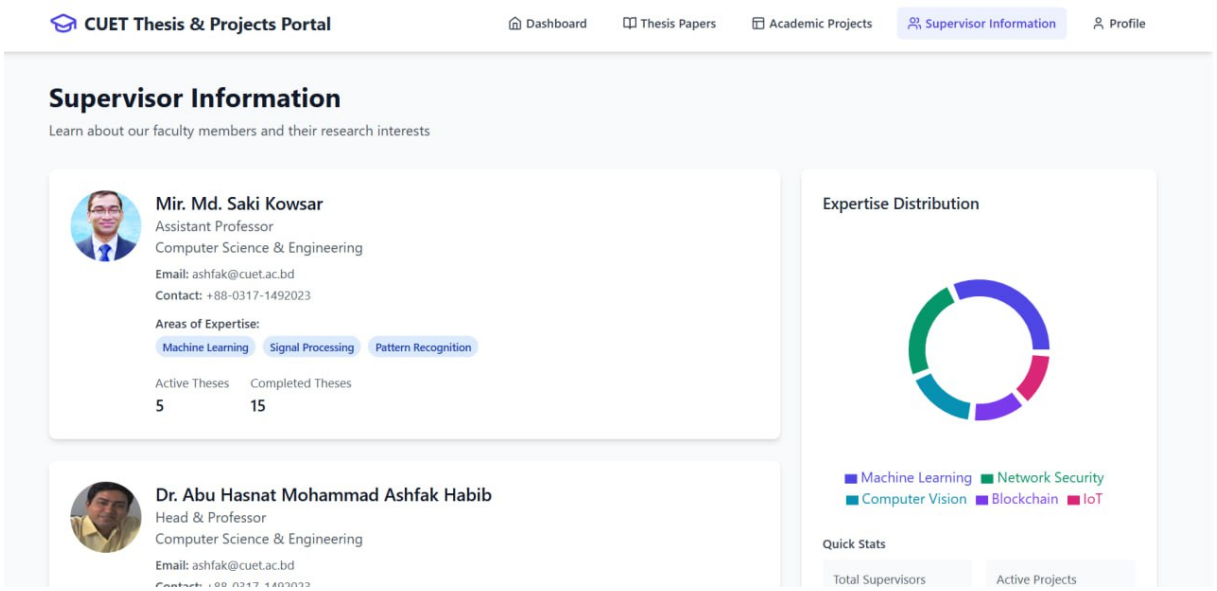


Figure 5.12: Supervisor Page

Project Page

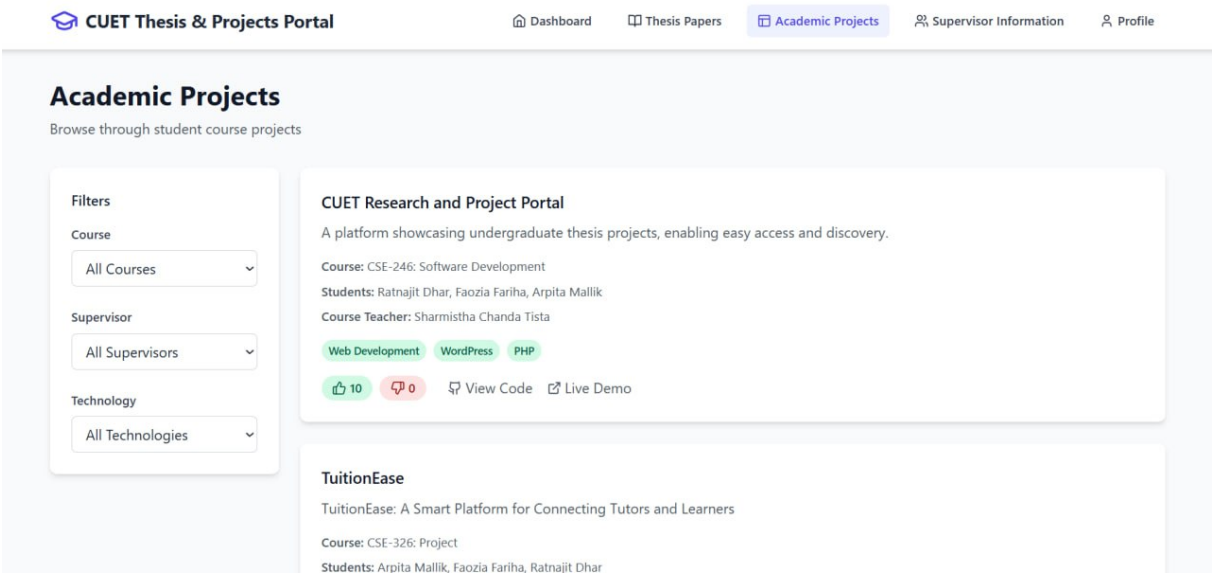


Figure 5.13: Project Page

Profile

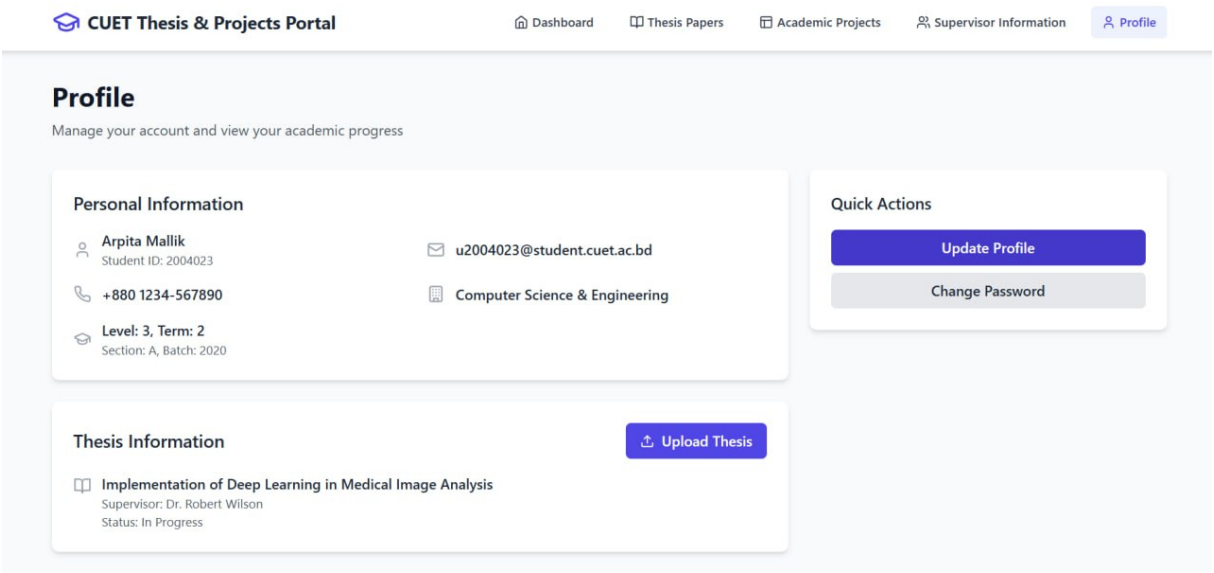


Figure 5.14: Profile

Testing and sustainability plan

6.1 Requirements/specifications-based system level test cases

In the software development process, components of a program undergo rigorous examination through component testing. In this phase, individual components are scrutinized in isolation, without being integrated with any other component. Component testing allows for the identification of issues specific to each part of the software, ensuring that each unit operates correctly and fulfills its designated functionality. The following tabulated representation outlines the requirements/specifications-based system level test cases:

Requirement ID	**Requirement Statement**	**Must/ Want**	**Comment**
R-Login-01	User must login to view dashboard and other pages	Must	N/A
R-Dashboard-01	Display the thesis information in the dashboard	Must	N/A
R-Thesis-01	Display & have download options for the thesis papers in the page	Must	N/A
R-Thesis-02	Filter the thesis papers based on their information	Must	N/A
R-Projects-01	Display & have downlaod options for Projects in the page	Must	N/A
R-Projects-02	Filter the projects based on their information	Must	N/A
R-Supervisor-01	Display Supervisor Information	Must	N/A
R-Profile-01	Update Profile Information	Must	N/A
R-Profile-02	Upload thesis & projects	Must	N/A

Table 6.1: Requirements/specifications-based statements for CUET Thesis & Project Portal

Project Name	CUET Thesis & Project Portal					
Module Name	User Login					
Created By	Arpita Mallik					
Reviewed By	Ratnajit Dhar, Faozia Fariha					
Date of Creation	14-01-25					
Date of Review	14-01-25					
Test Case ID	**Scenario**	**Steps**	**Test Data**	**Expected Result**	**Actual Result**	**Status**
TC-UL-01	Check user login to view dashboard	1. Go to the login page 2. Give user credentials 3. Click Login Button	login.tsx	User gets access to dashboard and other pages	User gets access to dashboard and other pages	Passed

Table 6.2: Requirements/specifications-based system level test cases (01)

Project Name	CUET Thesis & Project Portal					
Module Name	Thesis Pages					
Created By	Ratnajit Dhar					
Reviewed By	Arpita Mallik, Faozia Fariha					
Date of Creation	14-01-25					
Date of Review	17-01-25					
Test Case ID	**Scenario**	**Steps**	**Test Data**	**Expected Result**	**Actual Result**	**Status**
TC-TP-01	View all the thesis papers correctlty	1. After login, click on 'Thesis Page' on the navbar	thesis.tsx	User can view the thesis papers in a card box div with all its information	User can view the thesis papers in a card box div with all its information	Passed
TC-TP-02	View thesis papers	1. Click on View button 2. View thesis paper	thesis.tsx	User will redirect to view page	User redirected to the view page	Passed

Table 6.3: Requirements/specifications-based system level test cases (02)

Project Name	CUET Thesis & Project Portal					
Module Name	Academic Projects					
Created By	Faozia Fariha					
Reviewed By	Ratnajit Dhar, Arpita Mallik					
Date of Creation	14-01-25					
Date of Review	17-01-25					
Test Case ID	**Scenario**	**Steps**	**Test Data**	**Expected Result**	**Actual Result**	**Status**
TC-AP-01	View all the academic projects correctlty	1. After login, click on academic projects" on the navbar	projects.tsx	User can view the academic projects in a card box div with all its information	User can view the academic projects in a card box div with all its information	Passed
TC-AP-02	Download academic projects	1. View academic projects Page 2. Click on Download button	the-sis.tsx	User will redirect to a github repository	User will be redirected to a github repository	Passed

Table 6.4: Requirements/specifications-based system level test cases (03)

Project Name	CUET Thesis & Project Portal					
Module Name	Supervisor Page					
Created By	Arpita Mallik					
Reviewed By	Faozia Fariha, Ratnajit Dhar					
Date of Creation	17-01-25					
Date of Review	19-01-25					
Test Case ID	**Scenario**	**Steps**	**Test Data**	**Expected Result**	**Actual Result**	**Status**
TC-SP-01	View all the supervisor information	1. After login, click on supervisor page on the navbar	supervisor.tsx	User can view the informations of supervisor at one place	User can view the informations of supervisor at one place	Passed

Table 6.5: Requirements/specifications-based system level test cases (04)

Project Name	CUET Thesis & Project Portal					
Module Name	Supervisor Page					
Created By	Faozia Fariha					
Reviewed By	Arpita Mallik, Ratnajit Dhar					
Date of Creation	20-01-25					
Date of Review	21-01-25					
Test Case ID	**Scenario**	**Steps**	**Test Data**	**Expected Result**	**Actual Result**	**Status**
TC-PP-01	View the profile page	1. After login, click profile page on the navbar	pro-file.tsx	User can view the profile informations	User can view the profile informations	Passed

Table 6.6: Requirements/specifications-based system level test cases (05)

6.2 Application Program Interface(API)

Integration: Narratives

The CUET Thesis and Project Portal API enables seamless access to undergraduate thesis papers and course projects from the CSE department. It handles authentication, user roles, and content interactions efficiently. All endpoints have been tested using Postman to ensure reliability and security. This platform empowers students and faculty to explore research, provide feedback, and contribute to a growing academic repository.

6.2.1 Student Information

Purpose: Fetch Student Information

Request:

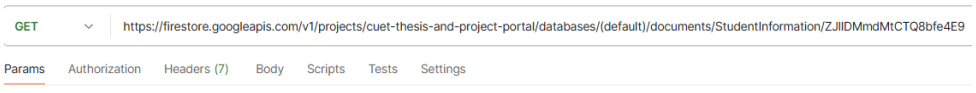


Figure 6.1: GET Request of Student Information

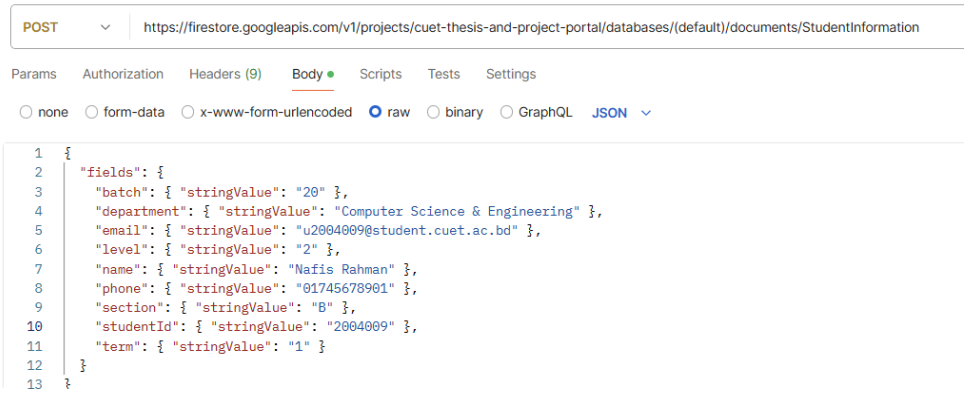


Figure 6.2: POST Request of Student Information

Response:

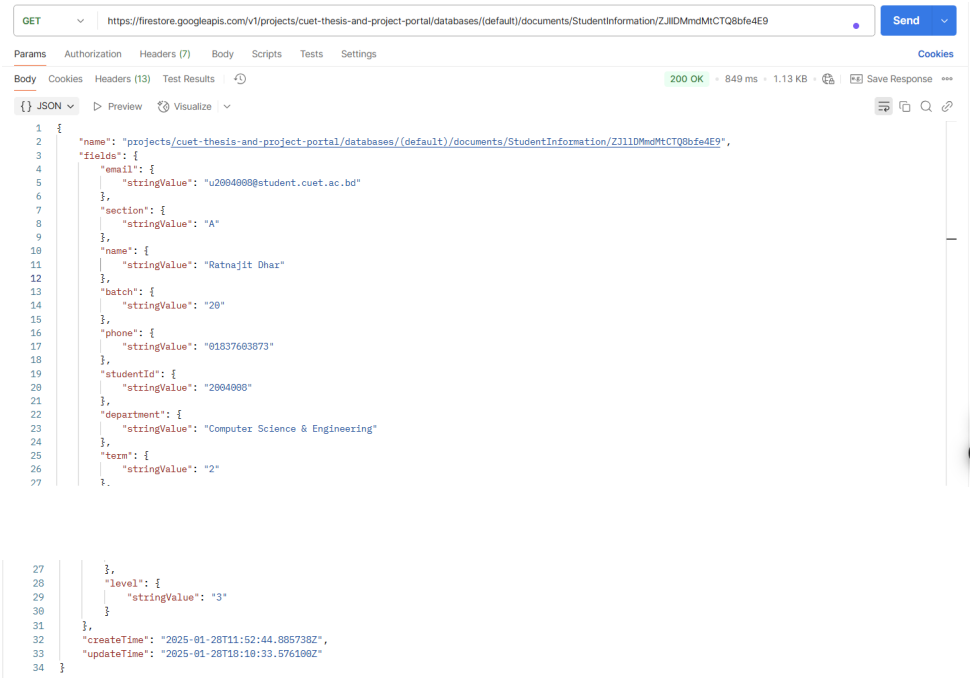


Figure 6.3: GET Response of Student Information

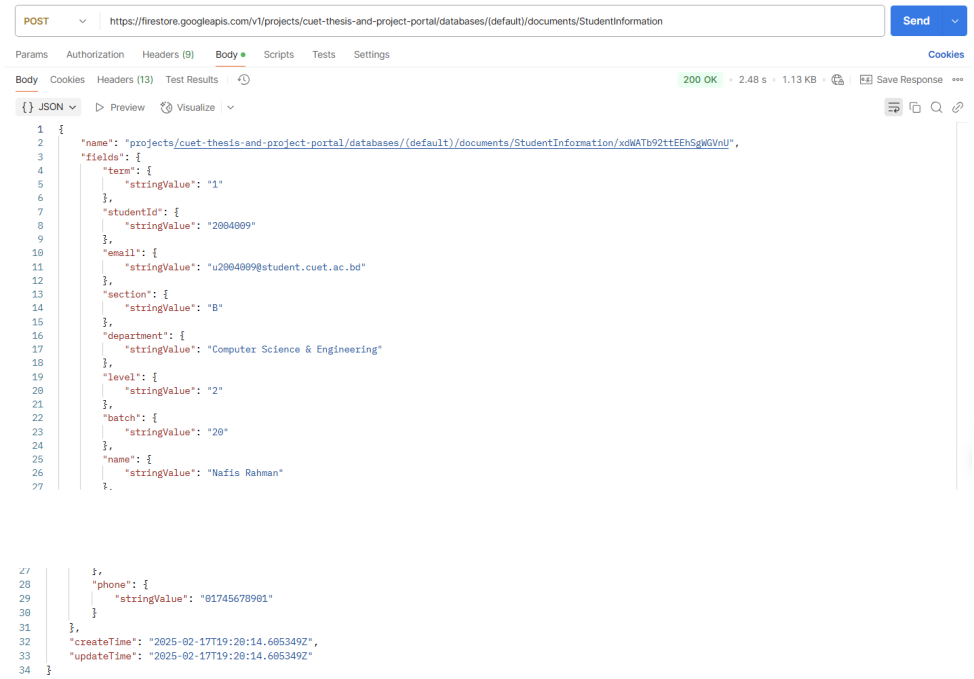


Figure 6.4: POST Response of Student Information

6.2.2 Supervisor Information

Purpose: Fetch Supervisor Information

Request:

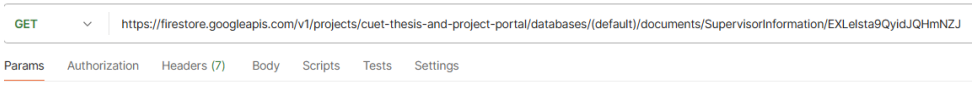


Figure 6.5: GET Request of Supervisor Information



Figure 6.6: POST Request of Supervisor Information

Response:

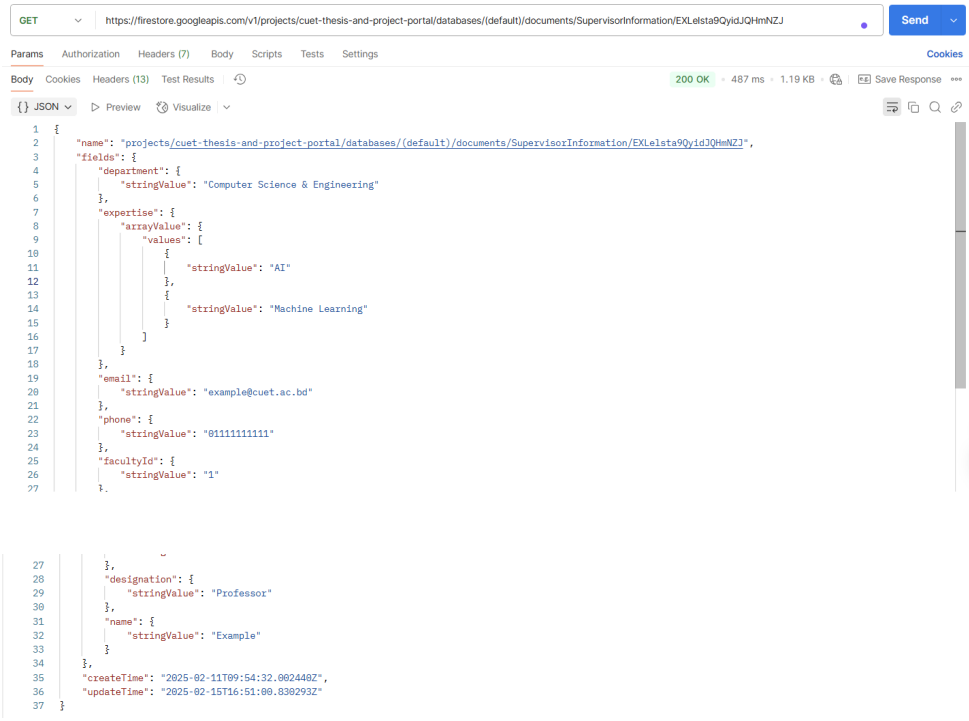


Figure 6.7: GET Response of Supervisor Information

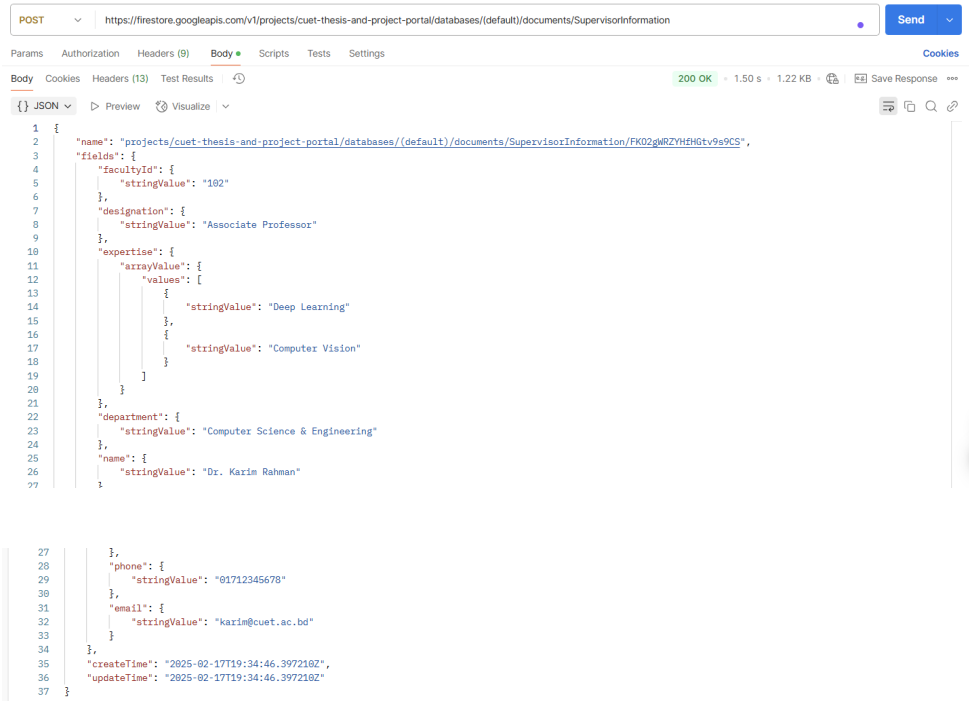


Figure 6.8: POST Response of Supervisor Information

6.2.3 Thesis Papers

Purpose: Fetch Thesis Information

Request:

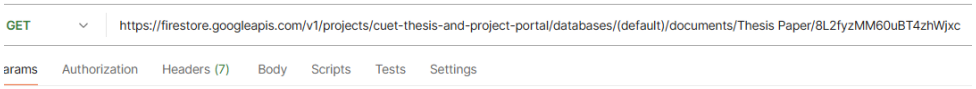


Figure 6.9: GET Request of Thesis Information

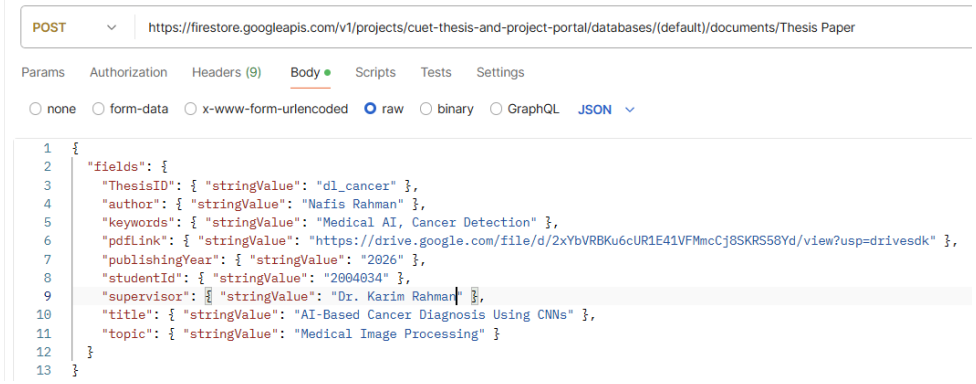


Figure 6.10: POST Request of Thesis Information

Response:

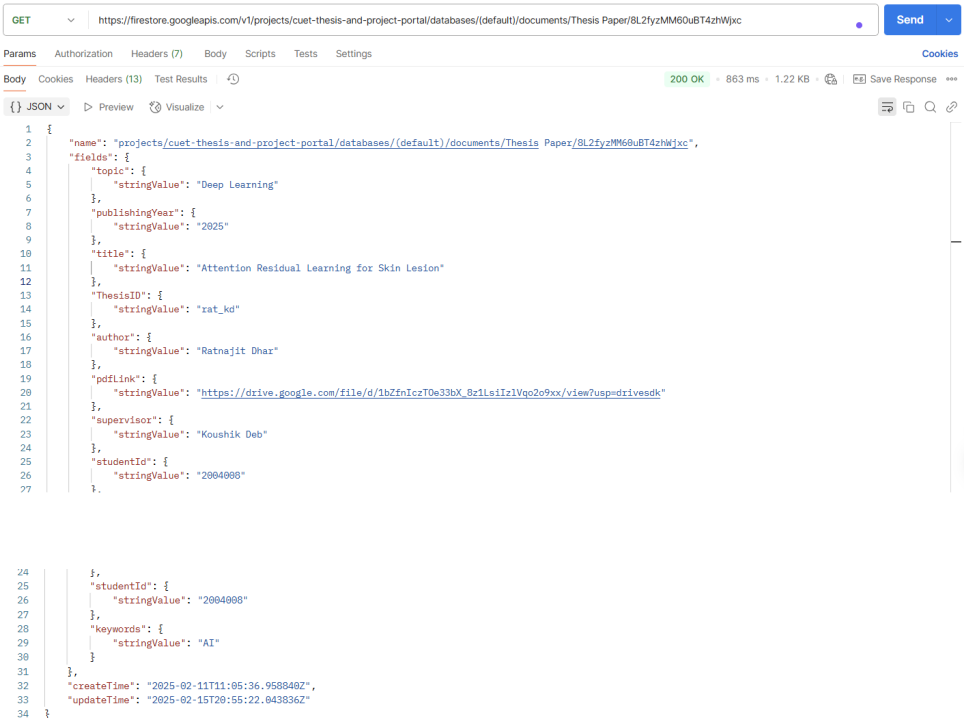


Figure 6.11: GET Response of Thesis Information

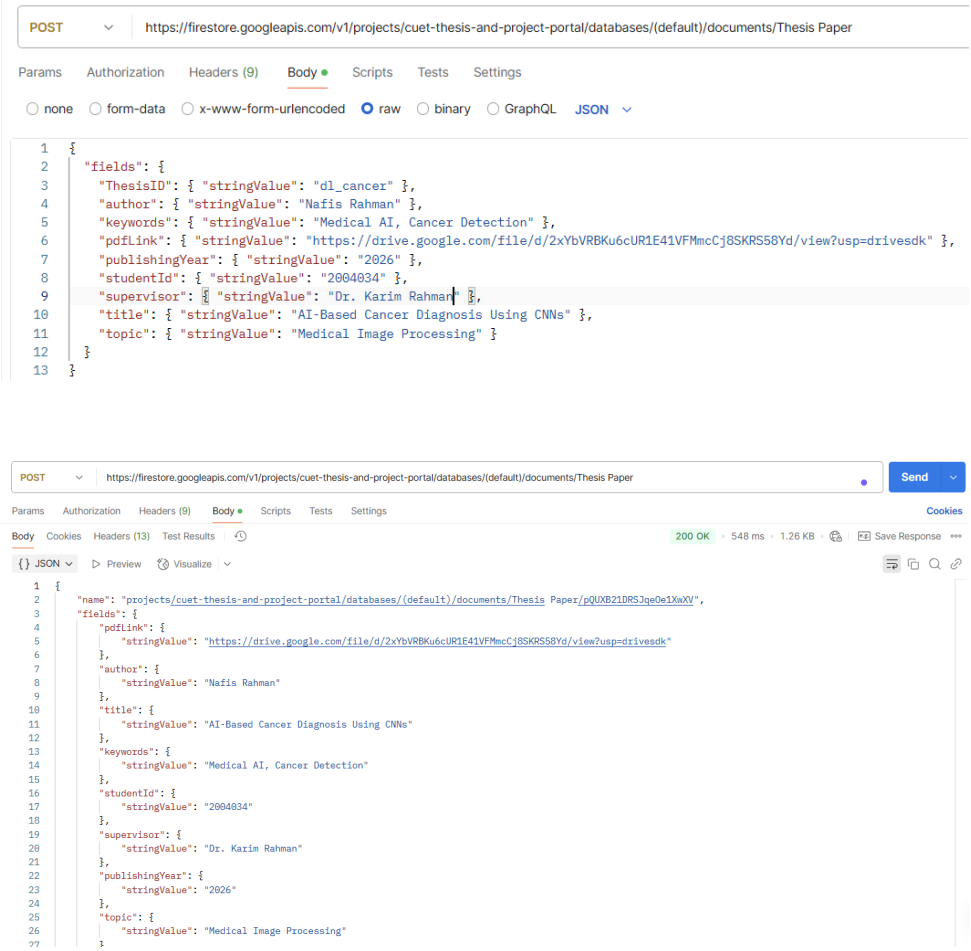


Figure 6.12: POST Response of Thesis Information

6.2.4 Academic Projects

Purpose: Fetch Project Information

Request:

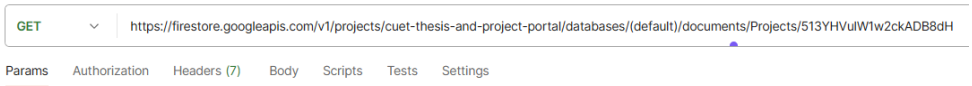


Figure 6.13: GET Request of Project Information

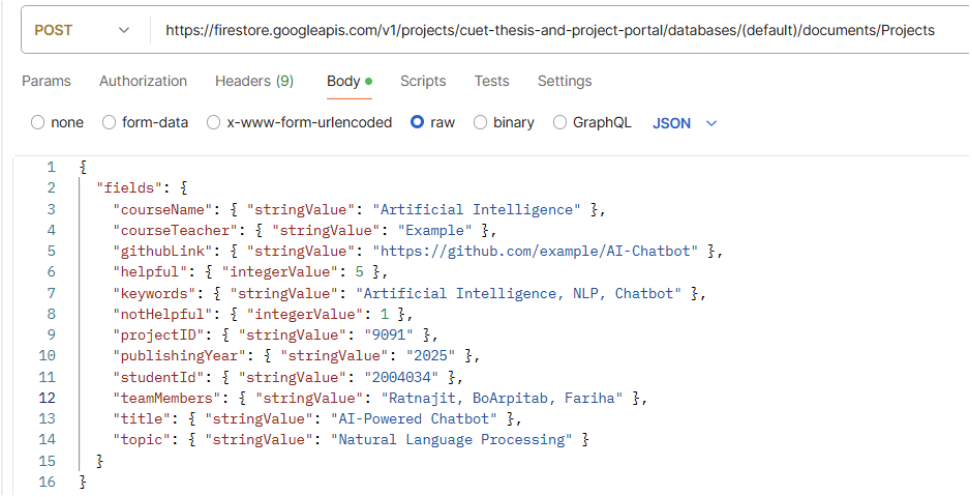


Figure 6.14: POST Request of Project Information

Response:

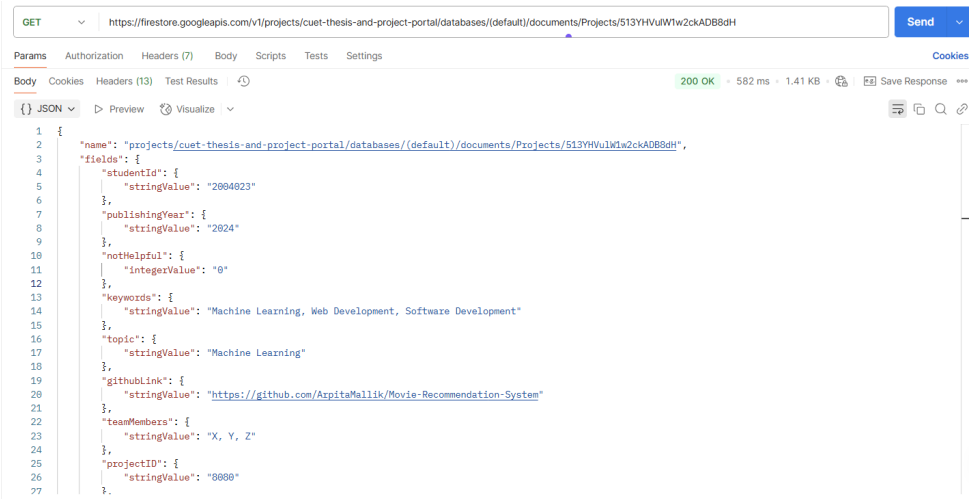


Figure 6.15: GET Response of Project Information

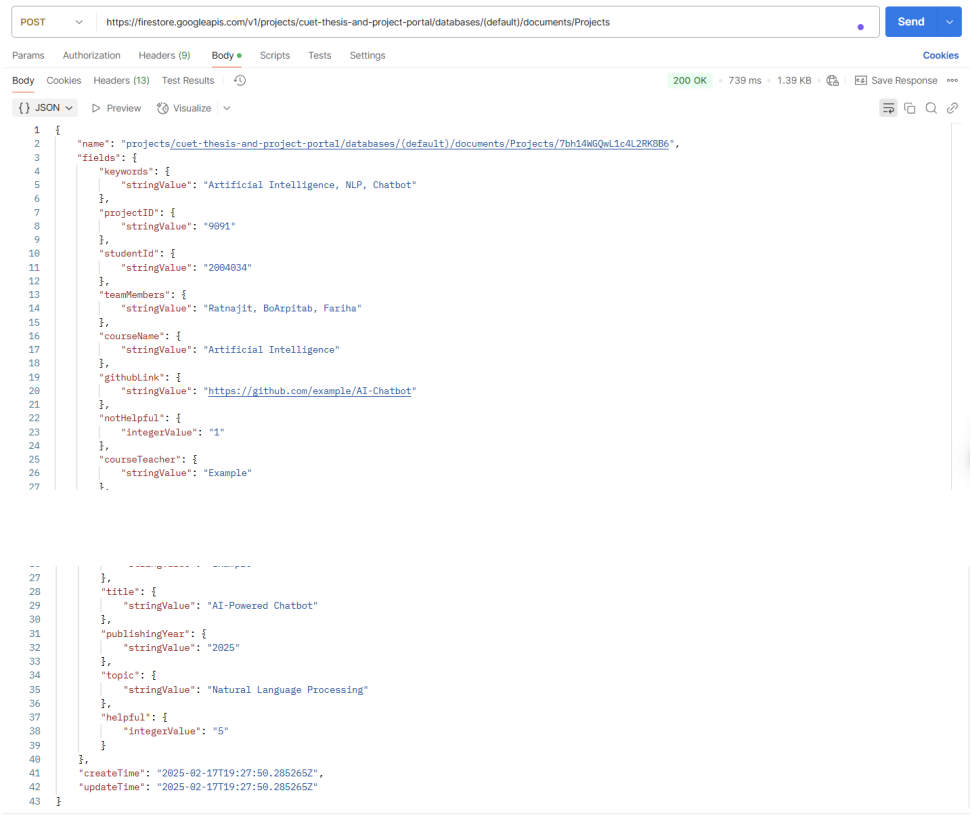


Figure 6.16: POST Response of Project Information

6.2.5 Pending Theses

Purpose: Fetch Pending Theses Information

Request:

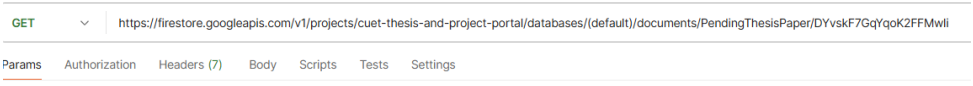


Figure 6.17: GET Request of Pending Theses

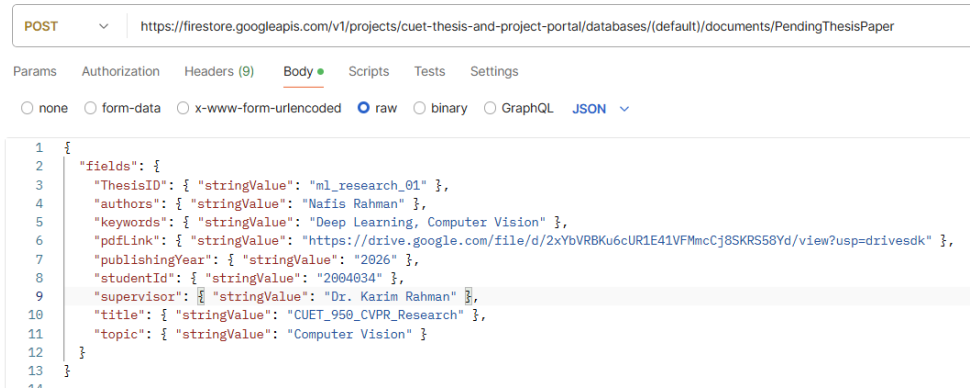


Figure 6.18: POST Request of Pending Theses

Response:

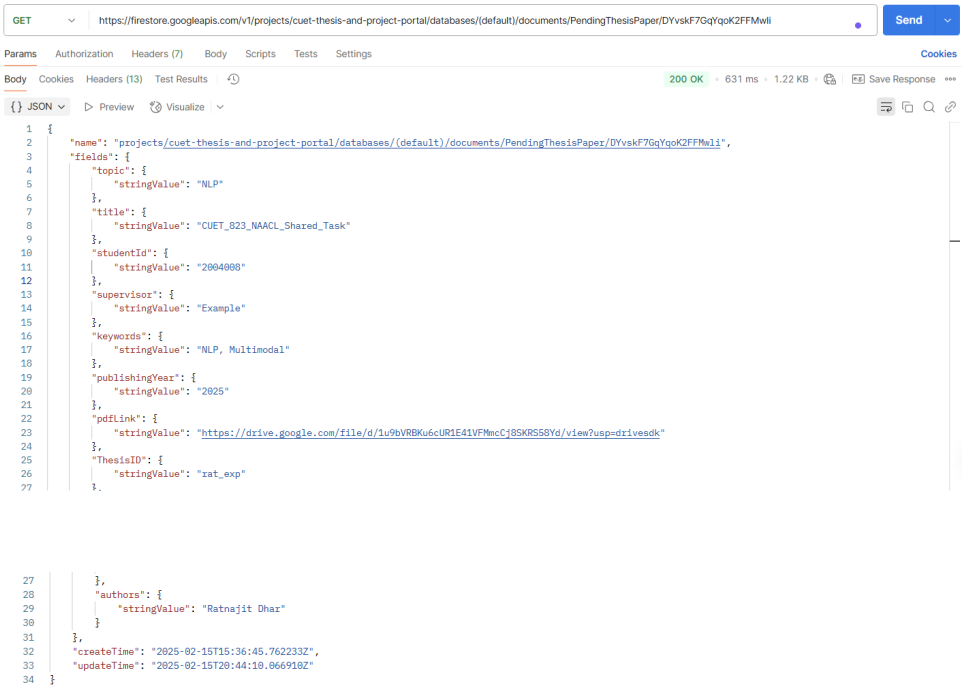


Figure 6.19: GET Response of Pending Theses

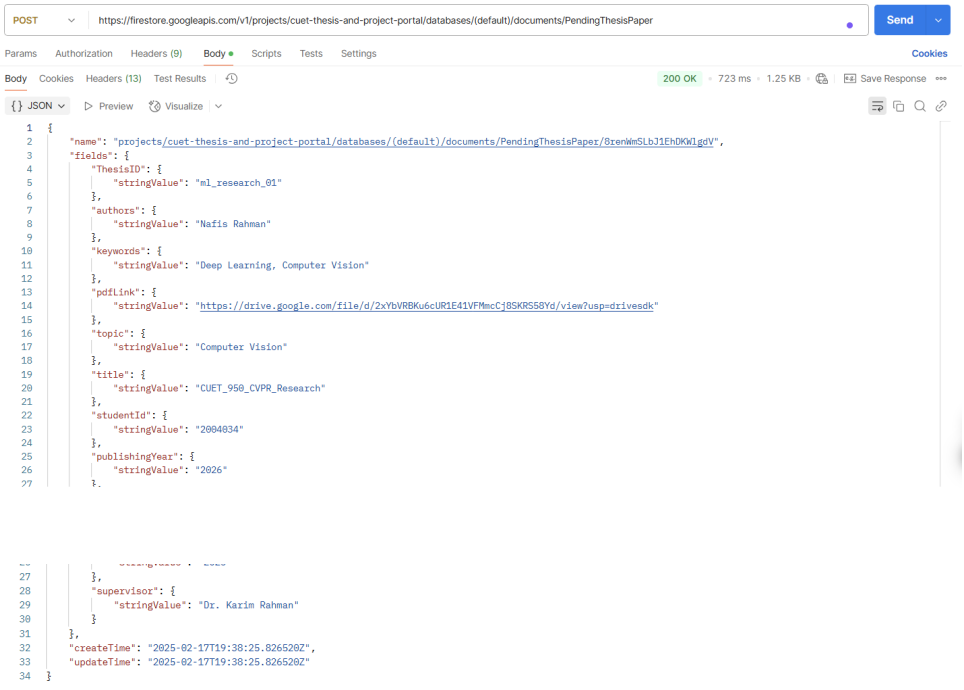


Figure 6.20: POST Response of Pending Theses

6.2.6 Pending Projects

Purpose: Fetch Pending Projects Information

Request:

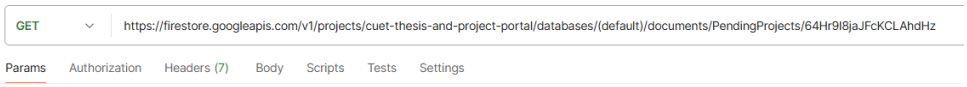


Figure 6.21: GET Request of Pending Projects

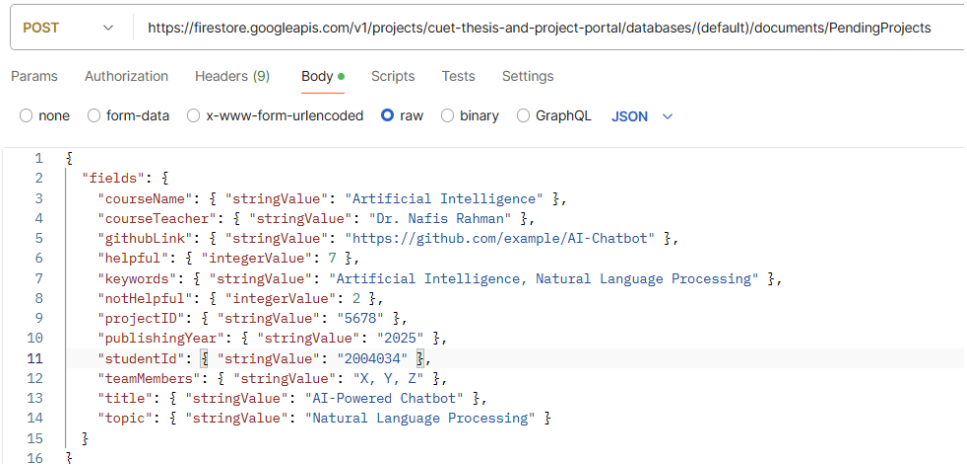


Figure 6.22: POST Request of Pending Projects

Response:

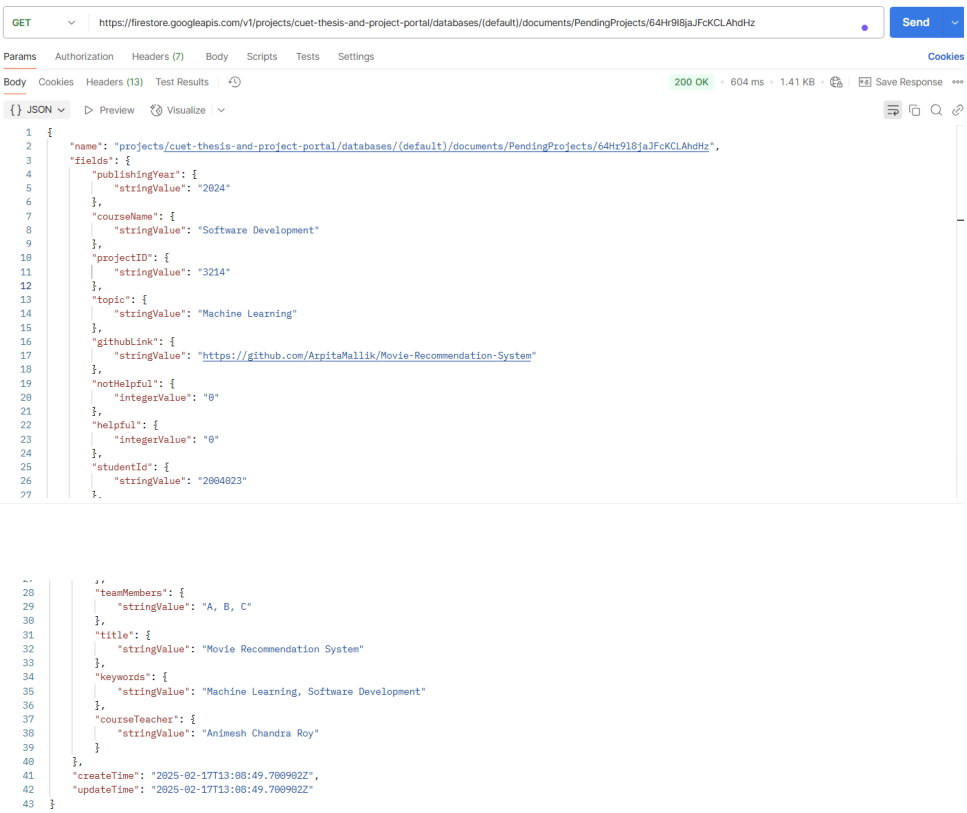


Figure 6.23: GET Response of Pending Projects

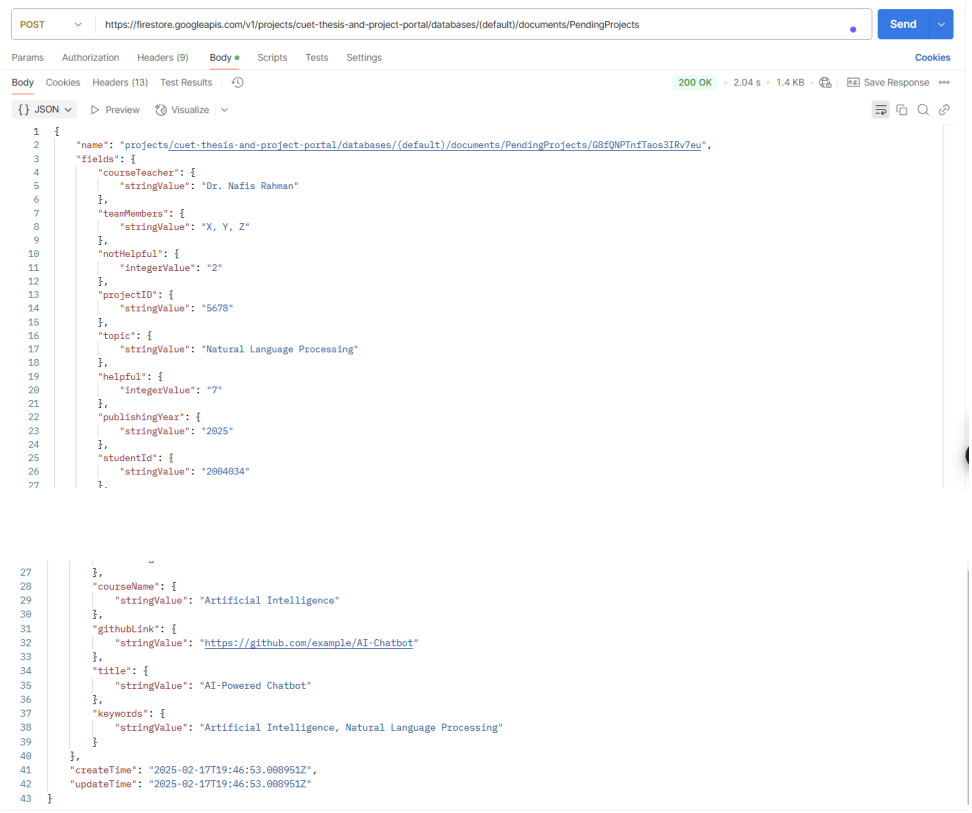


Figure 6.24: POST Response of Pending Projects

6.3 Assessment of the goodness of the test suite

Assessing the effectiveness of a test suite for the CUET Thesis and Project Portal involves evaluating its coverage, reliability, and ability to detect potential issues. The following key factors should be considered:

Test Coverage: Evaluate the comprehensiveness of the test suite in covering different functionalities and features of the CUET Thesis and Project Portal. This includes testing various modules, user interactions, and system behaviors. A well-structured test suite should ensure that a significant portion of the portal's functionality is covered to identify potential bugs and weaknesses effectively.

Functional Testing: Assess the thoroughness of functional testing by verifying how well essential features of the CUET Thesis and Project Portal, such as project submission, thesis management, faculty-student interactions, and document review processes, are tested. The test suite should incorporate typical user scenarios and edge cases to validate the portal's functionality under different conditions.

User Interface (UI) Testing: Verify if the test suite includes UI testing to ensure that the graphical user interface of the CUET Thesis and Project Portal functions correctly across different browsers and devices. This includes checking layout consistency, responsiveness, and usability to provide an optimal user experience for students, faculty members, and administrators.

Performance Testing: Evaluate whether the test suite includes performance testing to assess the responsiveness and scalability of the CUET Thesis and Project Portal under

various loads and network conditions. Performance testing should identify bottlenecks such as slow response times or resource inefficiencies and provide insights for optimizing the portal's performance.

Security Testing: Check if the test suite incorporates security testing to detect potential vulnerabilities and security risks within the CUET Thesis and Project Portal. Security assessments should include authentication mechanisms, data encryption, and protection against common threats such as SQL injection, cross-site scripting (XSS), and unauthorized data access. Strengthening security measures will help ensure the confidentiality and integrity of thesis and project-related data.

By thoroughly evaluating these factors, the effectiveness of the test suite for the CUET Thesis and Project Portal can be ensured, leading to a robust, reliable, and secure platform for students and faculty members.

6.4 Sustainability Plan

The factors contributing to the sustainability of the CUET Thesis and Project Portal collectively form its sustainability strategy. This strategy aims to support continuous growth, optimize efficiency, and balance immediate needs with long-term objectives, fostering sustainable development over time.

6.4.1 Scalability

Scalability is a crucial factor in evaluating the capability of the CUET Thesis and Project Portal for growth and improvement. A scalable system should efficiently handle varying user loads, from a few users to large research communities. Key considerations for scalability include:

- **Performance Optimization:** The CUET Thesis and Project Portal is designed to efficiently manage increased user traffic and research data submissions. Performance optimizations, such as streamlined algorithms and server-side caching, ensure smooth operation even as more users and projects are added.
- **Cloud Infrastructure:** Utilizing cloud infrastructure allows the CUET Thesis and Project Portal to scale horizontally by adding more resources, such as computing power and storage, as needed. Cloud services provide flexibility to accommodate varying demands and ensure consistent performance.
- **Load Balancing:** Implementing load balancing mechanisms distributes incoming traffic evenly across multiple servers, preventing overload and ensuring stable performance, especially during peak submission deadlines and research activities.

6.4.2 Flexibility / Customization

- **Plugin and Extension Support:** The CUET Thesis and Project Portal can support plugins or extensions, allowing users to enhance their experience by integrating additional features or research tools tailored to their needs.
- **Cross-Browser Compatibility:** Ensuring seamless operation across various web

browsers provides flexibility to users with different preferences, maintaining a consistent experience across platforms.

- **User Preferences and Settings:** Enabling users to personalize their portal experience by configuring settings such as default display options, themes, or notification preferences enhances usability and accessibility.
- **Real-time Collaboration:** Implementing real-time collaboration features within the CUET Thesis and Project Portal facilitates simultaneous participation and interaction among students and faculty. This enhances teamwork, improves research coordination, and fosters an engaging academic community.

Acknowledgement

We express our sincere gratitude for the dedicated efforts and valuable time generously contributed by our course teachers:

Mir. Md. Saki Kowsar
Assistant Professor, Department of Computer Science and Engineering, CUET

Md. Al-Mamun Provath
Lecturer, Department of Computer Science and Engineering, CUET

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